

UFS Memory Device Data Sheet

Revision 1.00
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KIOXIA Corporation

256GB THGJFGT1E45BAIP**Table of Contents**

1. INTRODUCTION	3
2. FEATURES.....	5
2.1. UFS modules Interface.....	5
2.2. Pin Assignment and Description.....	5
2.3. Part Number	7
2.4. Operating Temperature	7
2.5. Storage Temperature.....	7
2.6. Performance.....	7
2.7. Power Supply.....	8
2.8. Product Architecture	8
3. PRODUCT SPECIFICATIONS	9
3.1. Package Dimensions.....	9
3.2. Marking Information	10
3.3. Density Specifications	10
3.4. M-PHY ^R Attributes.....	11
3.5. UniPro ^R Attributes	15
3.6. UPIU Transaction codes	21
3.7. UFS Descriptors	21
3.8. UFS Flags.....	32
3.9. UFS Attributes.....	33
3.10. UFS SCSI Commands	35
3.11. Mode Pages.....	35
3.12. Standard INQUIRY data.....	37
3.13. Vital Product Data.....	38
4. ELECTRICAL CHARACTERISTICS	43
4.1. Absolute Maximum Ratings.....	43
4.2. H/W Reset Operation.....	43
4.3. Reference Clock	44
4.4. Power Supply.....	45
4.5. Power mode control	47
5. Reliability Guidance.....	48
6. Reference	49
7. Document Revision History	49

1. INTRODUCTION

THGJFGT1E45BAIP is JEDEC UFS specification Ver.3.1 compliant NAND flash memory housed in 153 ball BGA package. This memory utilizes advanced NAND flash memory and a controller chip assembled as Multi Chip Module.

Figure 1 shows the system model of UFS Ver.3.1 memory. This memory supports HS-G1/G2/G3/G4 A/B GEAR, and PWM-G1/G2/G3/G4 GEAR, and has differential input / output pins, some functional pins, power pins and ground pins. Data line specification is defined in M-PHY^R, and reference clock and reset signal specification is defined in UFS.

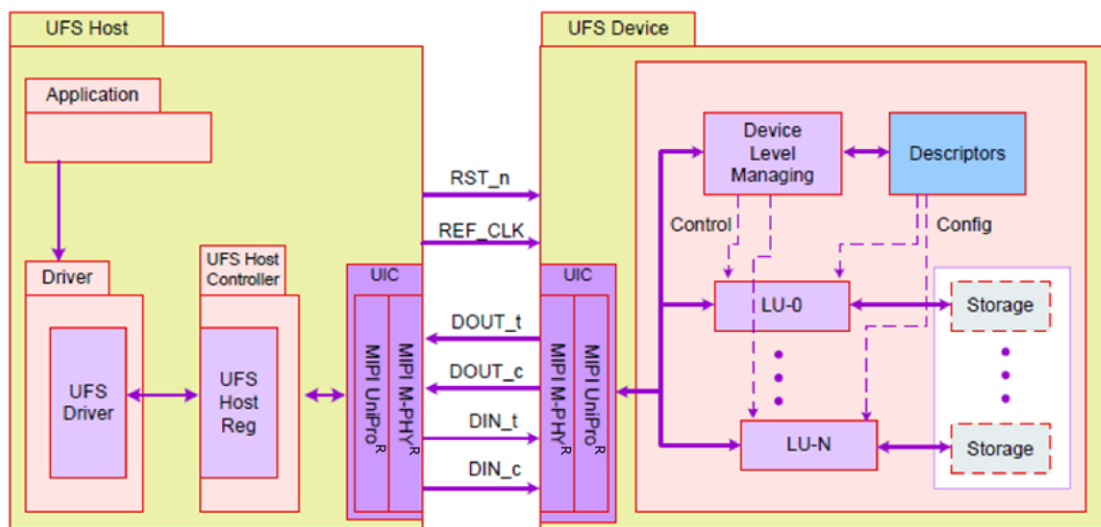


Figure 1 UFS System Model

Figure 2 shows UFS top level architecture. UFS adopts layered communication architecture based on a SCSI SAM architectural model. Lower layer specification than UTP is defined in MIPI UniPro^R and M-PHY^R specification.

A UFS module can contain one or more Logical Units (LUs), and LUs can be accessed simultaneously. THGJFGT1E45BAIP supports maximum 32 LUs. All access to UFS storage should be through correspondent LUs with UFS SCSI command subset, thus users need to configure appropriate LUs on UFS Configuration Descriptor for their own application. Device Descriptors, which includes device configuration variables, can be accessed through UFS Device Manager.

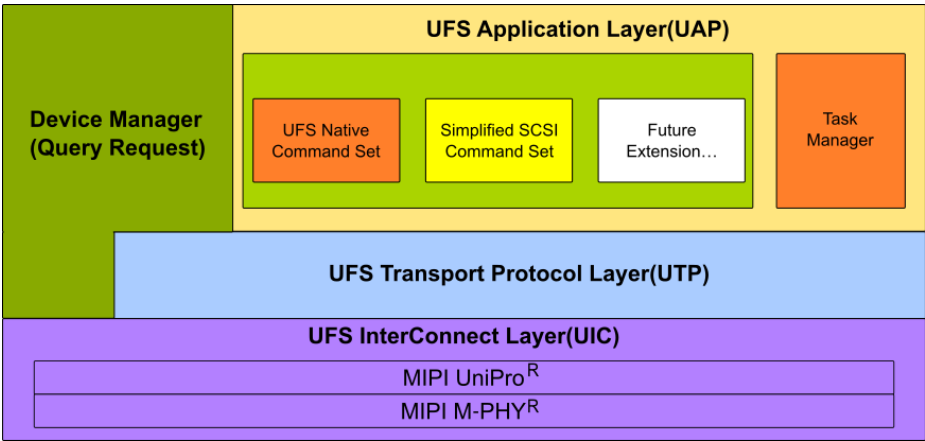


Figure 2 UFS Top Level Architecture

2. FEATURES

2.1. UFS modules Interface

UFS modules have downstream data lanes and upstream data lanes which are specified by M-PHY^R, reference clock and reset signal which are specified by UFS.

2.2. Pin Assignment and Description

P-WFBGA153-1113-0.50-702 (11mm x 13mm, H0.8mm(max.) package)

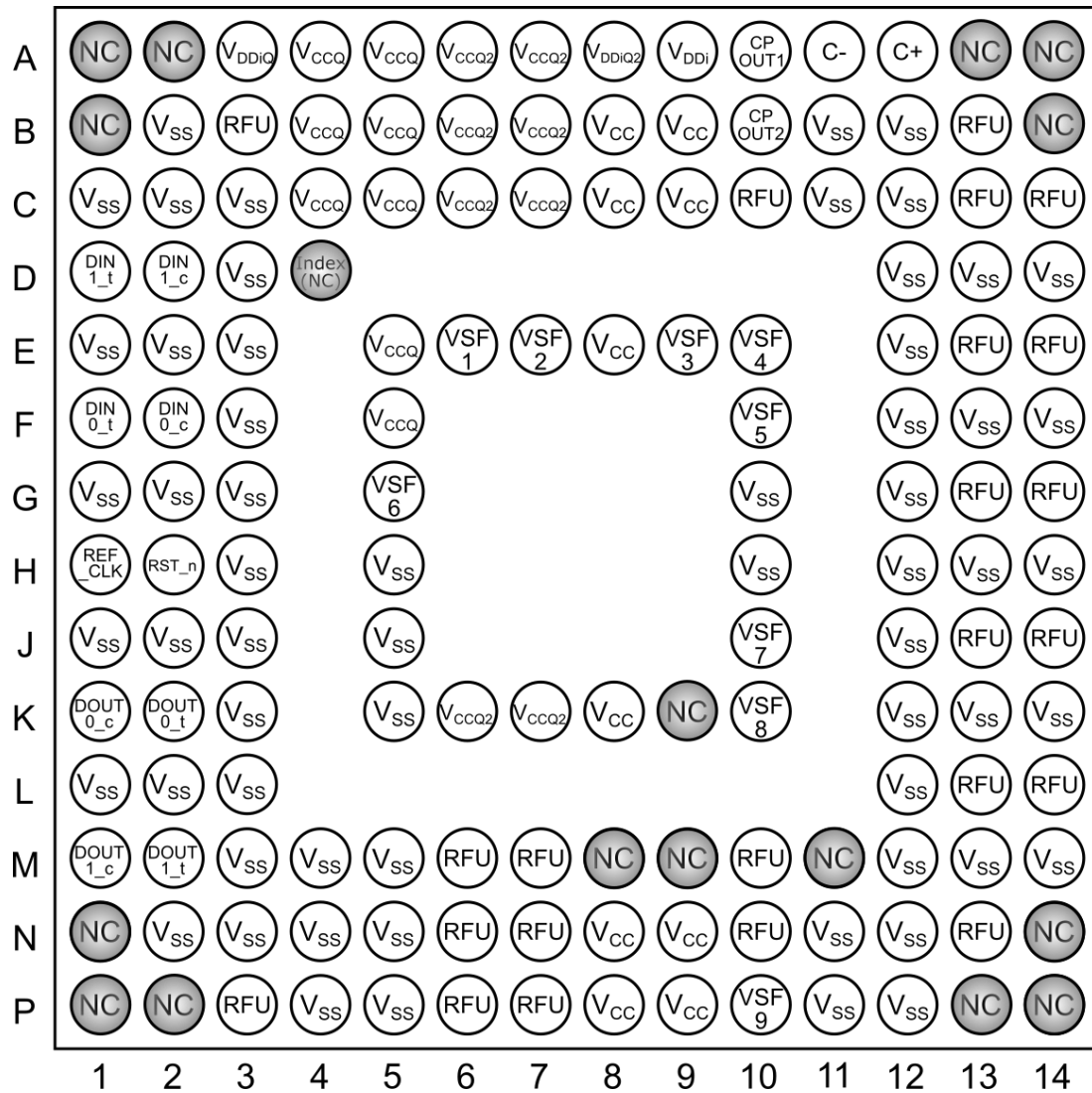


Figure 3 Pin Assignment (top view)

Table 1 Pin Description

Name	Type	Pin Number	Description
V _{CC}	Supply	B8-9, C8-9, E8, K8, N8-9, P8-9	Supply voltage for NAND chips
V _{CCQ}	Supply	A4-5, B4-5, C4-5, E5, F5	Supply voltage for a memory controller, NAND I/O interface and the M-PHY ^R interface
V _{CCQ2}	Supply	A6-7, B6-7, C6-7, K6-7	Not used
V _{DDI}	—	A9	V _{DDI} pin is internally connected to V _{CC}
V _{DDIQ}	Input	A3	Input terminal to provide bypass capacitor for V _{CCQ} internal regulator
V _{DDIQ2}	—	A8	Not used
V _{SS}	Supply	B2, B11-12, C1-3, C11-12, D3, D12-14, E1-3, E12, F3, F12-14, G1-3, G10, G12, H3, H5, H10, H12-14, J1-3, J5, J12, K3, K5, K12-14, L1-3, L12, M3-5, M12-14, N2-5, N11-12, P4-5, P11-12	Ground
RST_n ^{Note 1}	Input	H2	Input hardware reset signal
REF_CLK ^{Note 1}	Input	H1	Input reference clock When not active, this signal should be pull-down or driven low by the host SoC
DIN0_t	Input	F1	Downstream data lane 0
DIN0_c	Input	F2	Downstream data lane 0
DIN1_t	Input	D1	Downstream data lane 1
DIN1_c	Input	D2	Downstream data lane 1
DOUT0_t	Output	K2	Upstream data lane 0
DOUT0_c	Output	K1	Upstream data lane 0
DOUT1_t	Output	M2	Upstream data lane 1
DOUT1_c	Output	M1	Upstream data lane 1
CPOUT1	—	A10	Not used Shall be connected to ground or left floating
CPOUT2	—	B10	Not used Shall be connected to ground or left floating
C+	—	A12	Not used Shall be connected to ground or left floating
C-	—	A11	Not used Shall be connected to ground or left floating
VSN	—	E6-7, E9-10, F10, G5, J10, K10, P10	Shall be left floating
NC	—	A1, A2, A13, A14, B1, B14, D4, K9, M8, M9, M11, N1, N14, P1, P2, P13, P14	Shall be connected to ground or left floating
RFU	—	B3, B13, C10, C13, C14, E13, E14, G13, G14, J13, J14, L13, L14, M6, M7, M10, N6, N7, N10, N13, P3, P6, P7	Shall be left floating for future use

Note 1: Users need to input V_{CCQ} (1.14 to 1.26V) based signals to RST_n and REF_CLK to meet the electrical characteristics defined by UFS specification.

2.3. Part Number

Table 2 Product Part Number

Part Number	Density	NAND Flash Type	Package Size	Weight
THGJFGT1E45BAIP	256GB	4 x 512Gbit BiCS FLASH Gen.5	11mm x 13mm x 0.8mm(max.)	0.19g (typ.)

2.4. Operating Temperature

Table 3 Operating Temperature

Characteristics	min.	max.	Unit
Operating Temperature (T _C) Note 1, 4	-25 Note 2	+85 Note 3	°C

Note 1: Operating Temperature is the package case surface temperature of the UFS device when UFS device is operating.

Note 2: The minimum temperature should be based on the lowest temperature over the entire case surface.

Note 3: The maximum temperature should be based on the highest temperature over the entire case surface.

Note 4: Avoid locations where UFS devices may be exposed to water (wet, rain, dew condensation, etc.).

2.5. Storage Temperature

Table 4 Storage Temperature

Characteristics	min.	max.	Unit
Storage Temperature (T _C) Note 1, 4	-40 Note 2	+85 Note 3	°C

Note 1: Storage Temperature is the package case surface temperature of the UFS device when power is not supplied.

Note 2: The minimum temperature should be based on the lowest temperature over the entire case surface.

Note 3: The maximum temperature should be based on the highest temperature over the entire case surface.

Note 4: Avoid locations where UFS devices may be exposed to water (wet, rain, dew condensation, etc.).

2.6. Performance

HS-Gear with Rate B, Sequential Access (512KB access size)

Table 5 Performance

Parameter	HS-Gear	No. of Lane	Write Booster	Performance		Unit
				Read	Write	
Typical Performance	HS-G4	2	ON	2110	1450	MB/s
			OFF		525	

2.7. Power Supply

THGJFGT1E45BAIP supports 2 power supply of V_{CC} and V_{CCQ} and does not use V_{CCQ2} as power supply.

It is required to design layouts that supply below V_{CC} and V_{CCQ} voltage at the solder ball input terminals.

$V_{CC} = 2.7V$ to $3.6V$

$2.4V$ to $2.7V$

$V_{CCQ} = 1.14V$ to $1.26V$

2.8. Product Architecture

Figure 4 illustrates the main functional blocks of the UFS module, and Table 6 and Table 7 shows the recommended capacitor values for power supplies.

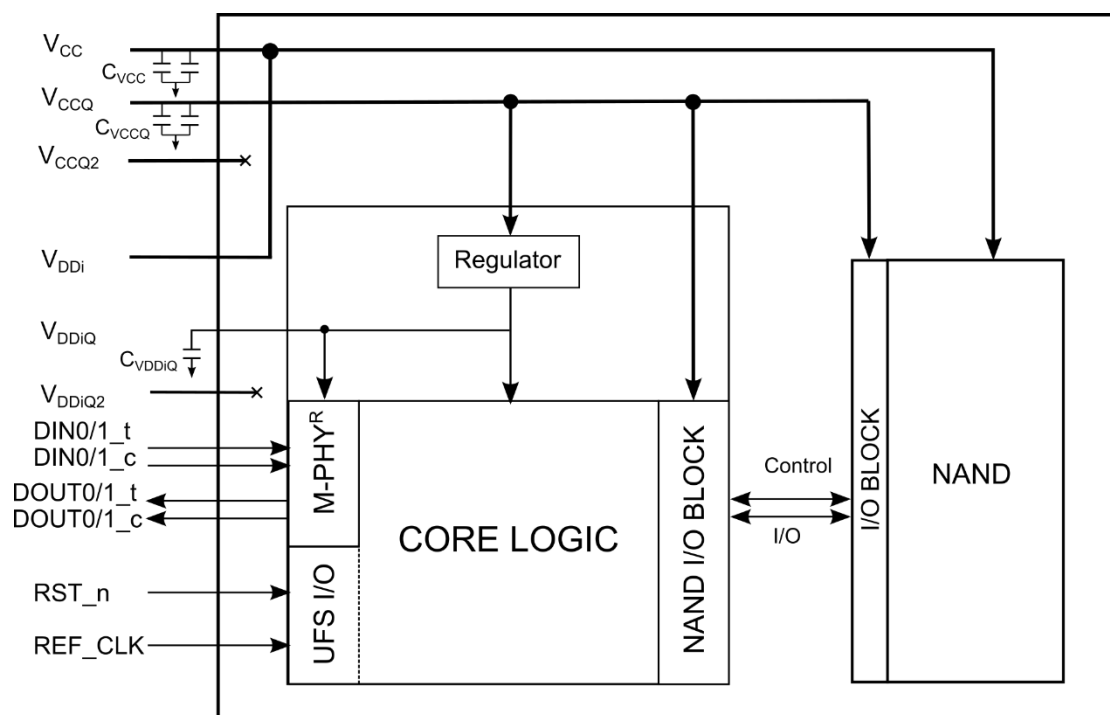


Figure 4 UFS module Block Diagram

Table 6 Required Capacitor Value for Power Supply

Parameter	Symbol	min.	typ.	max.	Unit
V_{DDiQ} capacitor	C_{VDDiQ}	0.7	2.2	2.4	μF

Table 7 Recommended Capacitor Value for Power Supply

Parameter	Symbol	Recommended Value	Unit
V_{CC} capacitor	C_{VCC}	4.7 and 0.1	μF
V_{CCQ} capacitor	C_{VCCQ}	2.2 and 0.1	μF

3. PRODUCT SPECIFICATIONS

3.1. Package Dimensions

P-WFBGA153-1113-0.50-702 (11mm x 13mm, H0.8mm(max.) package)

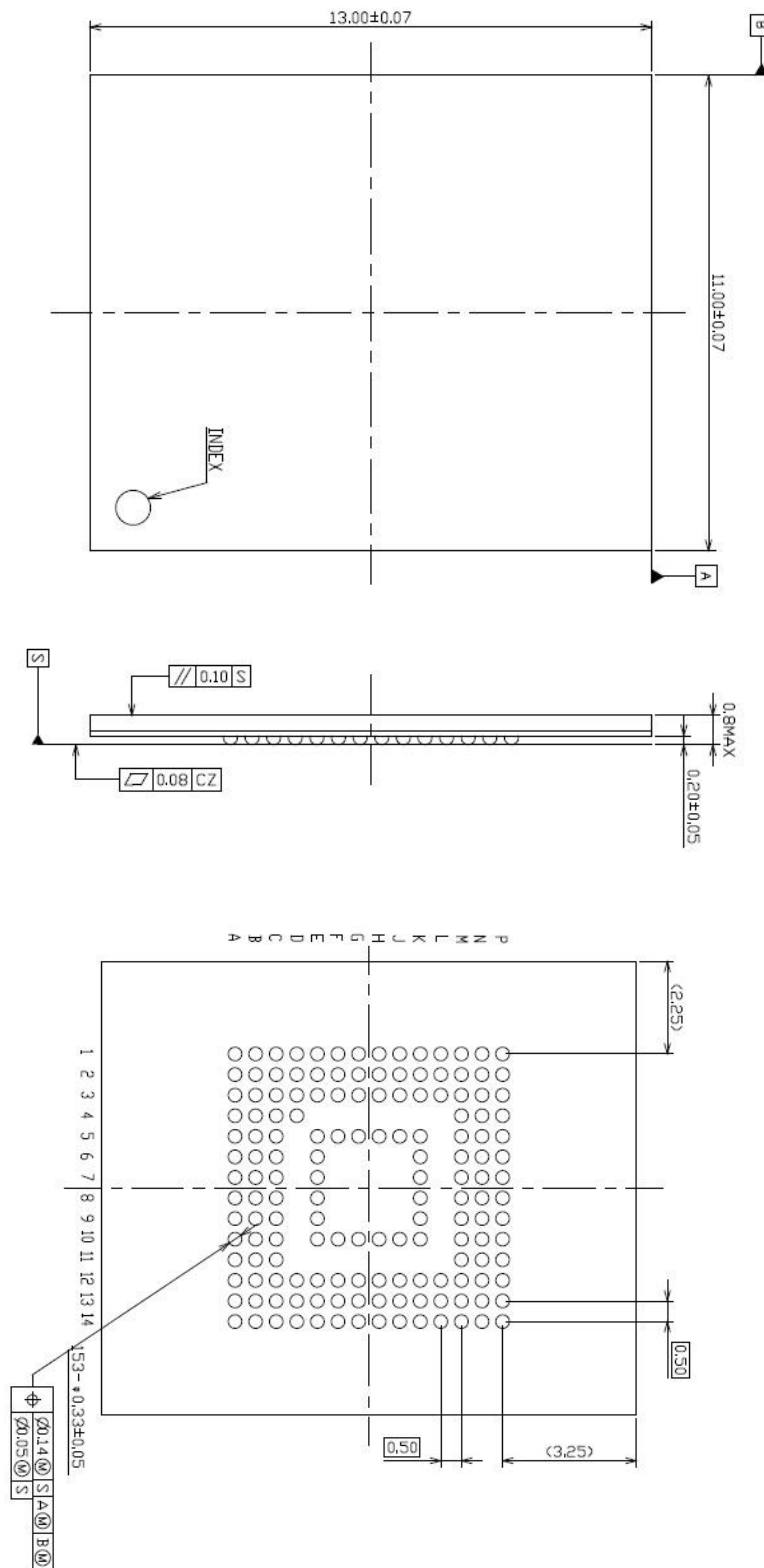


Figure 5 Package Dimensions (Unit: mm)

3.2. Marking Information

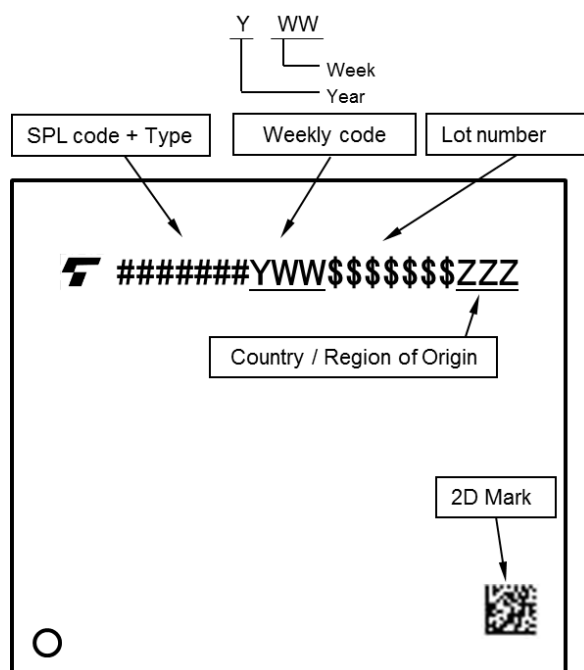


Figure 6 Marking Specification

Table 8 Marking Information

Part Number	Marking Information
THGJFGT1E45BAIP	##GT15PYWW\$\$\$\$\$ZZZ

3.3. Density Specifications

Table 9 User Area Density

Part Number	Density	User Area Density [Bytes]
THGJFGT1E45BAIP	256GB	255,944,818,688

Note: User area density is reduced if enhanced user data area is defined.

3.4. M-PHY^R Attributes

THGJFGT1E45BAIP supports 2 lanes. All M-PHY^R Attributes of each lane are identified by Selector number and Attribute ID.

Table 10 Lane Number and Selector Number

TX/RX	Selector
Lane 0 TX	0x0
Lane 1 TX	0x1
Lane 0 RX	0x4
Lane 1 RX	0x5

3.4.1. M-TX Capability Attributes

Table 11 M-TX Capability Attributes

Attribute Name	Access	Attribute ID	Reset Value ^{Note 1}	Notes
TX_HSMODE_Capability	R	0x01	0x01	Supports HS_MODE
TX_HSGEAR_Capability	R	0x02	0x04	Supports HS_G1 to HS_G4 (A/B)
TX_PWMG0_Capability	R	0x03	0x00	PWM_G0 is NOT supported
TX_PWMGEAR_Capability	R	0x04	0x04	Supports PWM_G1 to PWM_G4
TX_Amplitude_Capability	R	0x05	0x03	Supports Small and Large Amplitude
TX_ExternalSYNC_Capability	R	0x06	0x01	Supports external SYNC pattern
TX_HS_Unterminated_LINE_Drive_Capability	R	0x07	0x00	Not Supports un-terminated line in HS-MODE
TX_LS_Terminated_LINE_Drive_Capability	R	0x08	0x01	Supports terminated line in LS-MODE
TX_Min_SLEEP_NoConfig_Time_Capability	R	0x09	0x08	Minimum time in SLEEP state is 8SI
TX_Min_STALL_NoConfig_Time_Capability	R	0x0A	0x20	Minimum time in STALL state is 32SI
TX_Min_SAVE_Config_Time_Capability	R	0x0B	0xFA	Minimum reconfiguration time is 10μs
TX_REF_CLOCK_SHARED_Capability	R	0x0C	0x01	1: YES
TX_PHY_MajorMinor_Release_Capability	R	0x0D	0x41	Supports M-PHY ^R version 4.1
TX_PHY_Editorial_Release_Capability	R	0x0E	0x00	Revision 00
TX_Hibern8Time_Capability	R	0x0F	0x01	Minimum time in Hibern8 state is 100μs
TX_Advanced_Granularity_Capability	R	0x10	0x05	Supports fine granularity with the step size of 16μs
TX_Advanced_Hibern8Time_Capability	R	0x11	0x02	
TX_HS_Equalizer_Setting_Capability	R	0x12	0x03	Supports de-emphasis of 3.5dB and 6dB

Note 1: Reset value means the value after Link-up.

3.4.2. M-TX Configuration Attributes

Table 12 M-TX Configuration Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
TX_MODE	R ^{Note 3}	0x21	0x01	0x1: LS_MODE, 0x2: HS_MODE
TX_HSRATE_Series	R ^{Note 3}	0x22	0x01	0x1: Rate A Series, 0x2: Rate B Series
TX_HSGEAR	R ^{Note 3}	0x23	0x01	0x1: HS_G1, 0x2: HS_G2, 0x3: HS_G3, 0x4: HS_G4
TX_PWMGEAR	R ^{Note 3}	0x24	0x01	0x1: PWM_G1, 0x2: PWM_G2, 0x3: PWM_G3, 0x4: PWM_G4
TX_Amplitude	R/W	0x25	0x02	0x1: Small Amplitude, 0x2: Large Amplitude
TX_HS_SlewRate	R/W	0x26	0x00	Slew Rate control is NOT supported
TX_SYNC_Source	R/W	0x27	0x00	0: INTERNAL_SYNC, 1: EXTERNAL_SYNC
TX_HS_SYNC_LENGTH ^{Note 4}	R/W	0x28	—	Bit[7:6]: 0x0: FINE, 0x1: COARSE
				Bit[5:0]: Valid range: 0x00 - 0x0F This value is 0x4F before Link-up, and is updated during Link-up depending on the capability of host and device
TX_HS_PREPARE_LENGTH ^{Note 4}	R/W	0x29	—	Valid range: 0x6 - 0xF This value is 0x0F before Link-up, and is updated during Link-up depending on the capability of host and device
TX_LS_PREPARE_LENGTH ^{Note 4}	R/W	0x2A	—	Valid range: 0x9 - 0xF This value is 0x0A before Link-up, and is updated during Link-up depending on the capability of host and device
TX_HIBERN8_Control	R ^{Note 3}	0x2B	0x00	0: EXIT, 1: ENTER This value is 1 before Link-up
TX_LCC_Enable	R	0x2C	0x00	0: NO, 1: YES
TX_PWM_BURST_Closure_Extension	R/W	0x2D	—	Valid range: 0x00 - 0xFF This value is 0x20 before Link-up, and is updated during Link-up depending on the capability of host and device
TX_BYPASS_8B10B_Enable	R/W	0x2E	0x00	0: FALSE, 1: TRUE
TX_DRIVER_POLARITY	R/W	0x2F	0x00	0: NORMAL, 1: INVERTED
TX_HS_Unterminated_LINE_Drive_Enable	R ^{Note 3}	0x30	0x00	0: HS Terminated, 1: HS Un-terminated
TX_LS_Terminated_LINE_Drive_Enable	R ^{Note 3}	0x31	0x00	0: LS Un-terminated, 1: LS Terminated
TX_LCC_Sequencer	R ^{Note 3}	0x32	0x00	Bit[7]: LCC WRITE ATTRIBUTE 0: Not requested, 1: Requested
				Bit[6:3]: 0x0
				Bit[2]: LCC READ-VEND-INFO 0: Not requested, 1: Requested
				Bit[1]: LCC READ-MFG-INFO 0: Not requested, 1: Requested
TX_LCC_Sequencer	R ^{Note 3}	0x32	0x00	Bit[0]: LCC READ-CAPABILITY 0: Not requested, 1: Requested
TX_Min_ActivateTime	R/W	0x33	0x0F	Valid range: 0x00 - 0x0F

TX_Advanced_Granularity_Setting	R	0x35	0x00	Bit[2:1] step size b00: 4μs, b01: 8μs, b10: 16μs, b11: 32μs Bit[0] Supports advanced granularity 0: No, 1: Yes
TX_Advanced_Granularity	R	0x36	0x0F	Valid range: 0x1 - 0x0F
TX_HS_Equalizer_Setting	R/W	0x37	0x00	b000: No de-emphasis selected, b001: De-emphasis of 3.5dB selected, b010: De-emphasis of 6dB selected, b011 to b111: Reserved
Tx_Min_Sleep_NoConfig_Time	R/W	0x38	—	Invalid and not used in M-PHY ^R (optional setting)
Tx_Min_Stall_NoConfig_Time	R/W	0x39	—	Invalid and not used in M-PHY ^R (optional setting)
TX_HS_ADAPT_LENGTH	R/W	0x3A	0x91	ADAPT_Type: COARSE ADAPT_length: 17

Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: The attributes are write-protected by UniPro^R layer 1.8.

Note 4: The value has to be set equal or bigger than the value of RX side capability.

3.4.3. M-RX Capability Attributes

Table 13 M-RX Capability Attributes

Attribute Name	Access	Attribute ID	Reset Value ^{Note 1}	Notes
RX_HSMODE_Capability	R	0x81	0x01	Supports HS mode
RX_HSGEAR_Capability	R	0x82	0x04	Supports HS_G1 to HS_G4 (A/B)
RX_PWMG0_Capability	R	0x83	0x00	PWM_G0 is NOT supported
RX_PWMGEAR_Capability	R	0x84	0x04	Supports PWM_G1 to PWM_G4
RX_HS_Unterminated_Capability	R	0x85	0x00	Not Supports un-terminated line in HS-MODE
RX_LS_Terminated_Capability	R	0x86	0x01	Supports terminated line in LS-MODE
RX_Min_SLEEP_NoConfig_Time_Capability	R	0x87	0x0F	Minimum time in SLEEP state is 15SI
RX_Min_STALL_NoConfig_Time_Capability	R	0x88	0x70	Minimum time in STALL state is 112SI
RX_Min_SAVE_Config_Time_Capability	R	0x89	0xFA	Minimum reconfiguration time is 10μs
RX_REF_CLOCK_SHARED_Capability	R	0x8A	0x01	1: YES
RX_HS_G1_SYNC_LENGTH_Capability	R	0x8B	0x47	SYNC_range: COARSE, SYNC_length: 7
RX_HS_G1_PREPARE_LENGTH_Capability	R	0x8C	0x0F	HS_G1 PREPARE length is 15
RX_LS_PREPARE_LENGTH_Capability	R	0x8D	0x0A	PWM-BURST PREPARE length is 10 Reset value of 10 at a host side is receivable before capability exchange
RX_PWM_Burst_Closure_Length_Capability	R	0x8E	0x1F	Minimum burst closure time is 31SI
RX_Min_ActivateTime_Capability	R	0x8F	0x02	Minimum activate time is 200μs
RX_PHY_MajorMinor_Release_Capability	R	0x90	0x41	Supported M-PHY ^R version 4.1
RX_PHY_Editorial_Release_Capability	R	0x91	0x00	Revision 00
RX_Hibern8Time_Capability	R	0x92	0x01	Minimum time in HIBERN8 state is 100μs
RX_HS_G2_SYNC_LENGTH_Capability	R	0x94	0x48	SYNC_range: COARSE, SYNC_length: 8

RX_HS_G3_SYNC_LENGTH_Capability	R	0x95	0x49	SYNC_range: COARSE, SYNC_length: 9
RX_HS_G2_PREPARE_LENGTH_Capability	R	0x96	0x0F	HS_G2 PREPARE length is 15
RX_HS_G3_PREPARE_LENGTH_Capability	R	0x97	0x0F	HS_G3 PREPARE length is 15
RX_Advanced_Granularity_Capability	R	0x98	0x04	
RX_Advanced_Hibern8Time_Capability	R	0x99	0x02	
RX_Advanced_Min_ActivateTime_Capability	R	0x9A	0x0D	
RX_HS_G4_SYNC_LENGTH_Capability	R	0x9B	0x4A	SYNC_range: COARSE SYNC_length: 10
RX_HS_G4_PREPARE_LENGTH_Capability	R	0x9C	0x0F	HS_G4 PREPARE length multiplier for M-RX: 15
RX_HS_Equalizer_Setting_Capability	R	0x9D	0x01	Bit [0]: RX_HS_Equalizer_Setting_Capability: 0: NO equalization 1: YES equalization
RX_HS_ADAPT_REFRESH_Capability	R	0x9E	0x7F	ADAPT_Type: FINE ADAPT_length: 127
RX_HS_ADAPT_INITIAL_Capability	R	0x9F	0x7F	ADAPT_Type: FINE ADAPT_length: 127

Note 1: Reset value means the value after Link-up.

3.4.4. M-RX Configuration Attributes

Table 14 M-RX Configuration Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
RX_MODE	R ^{Note 3}	0xA1	0x01	0x01: LS_MODE, 0x02: HS_MODE
RX_HSRATE_Series	R ^{Note 3}	0xA2	0x01	0x01: Rate A Series 0x02: Rate B Series
RX_HSGEAR	R ^{Note 3}	0xA3	0x01	0x01: HS_G1, 0x02: HS_G2, 0x03: HS_G3, 0x04: HS_G4
RX_PWMGEAR	R ^{Note 3}	0xA4	0x01	0x01: PWM_G1, 0x02: PWM_G2, 0x03: PWM_G3, 0x04: PWM_G4
RX_LS_Terminated_Enable	R ^{Note 3}	0xA5	0x00	0: LS Un-terminated 1: LS Terminated
RX_HS_Unterminated_Enable	R ^{Note 3}	0xA6	0x00	0: HS Terminated 1: HS Un-terminated
RX_Enter_HIBERN8	R ^{Note 3}	0xA7	0x00	0: NO, 1: YES This value is 1 before Link-up
RX_BYPASS_8B10B_Enable	R/W	0xA8	0x00	0: Disabled, 1: Enabled
RX_Termination_Force_Enable	R/W	0xA9	0x00	0: NO, 1: YES
RX_ADAPT_Control	R/W	0xAA	0x01	ADAPT_type: INITIAL ADAPT_control: SYNC
RX_RECEIVER_POLARITY	R/W	0xAB	0x00	0: NORMAL 1: INVERTED

Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: The attributes are write-protected by UniPro^R layer 1.8.

3.4.5. M-TX & M-RX Status Attributes

Table 15 M-TX & M-RX Status Attributes

Attribute Name	Access	Attribute ID	Reset Value ^{Note 1}	Notes
TX_FSM_State	R	0x41	—	0x0: DISABLED, 0x1: HIBERN8 0x2: SLEEP, 0x3: STALL
RX_FSM_State	R	0xC1	—	0x4: LS-BURST, 0x5: HS-BURST 0x6: LINE-CFG, 0x7: LINE-RESET

Note 1: Reset value means the value after Link-up.

3.5. UniPro^R Attributes

3.5.1. UniPro^R L1.5 Attributes

Table 16 UniPro^R L1.5 Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
PA_PHY_Type	R	0x1500	0x01	M-PHY ^R
PA_AvailTxDataLanes	R	0x1520	0x2	Available TX Data Lane is 2
PA_AvailRxDataLanes	R	0x1540	0x2	Available RX Data Lane is 2
PA_MinRxTrailingClocks	R	0x1543	0x88	The host should set own PA_TxTrailingClocks to bigger value than this value
PA_TxHsG1SyncLength	R/W	0x1552	—	This value is 0x4F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxHsG1PrepareLength	R/W	0x1553	—	This value is 0x0F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxHsG2SyncLength	R/W	0x1554	—	This value is 0x4F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxHsG2PrepareLength	R/W	0x1555	—	This value is 0x0F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxHsG3SyncLength	R/W	0x1556	—	This value is 0x4F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxHsG3PrepareLength	R/W	0x1557	—	This value is 0x0F before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxMk2Extension	R/W	0x155A	—	This value is 0x00 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_PeerScrambling	R/W	0x155B	—	This value is 0x00 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxSkip	R/W	0x155C	—	This value is 0x00 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxSkipPeriod	R/W	0x155D	0xFA	
PA_Local_TX_LCC_Enable	R/W	0x155E	0x00	
PA_Peer_TX_LCC_Enable	R/W	0x155F	—	This value is 0x01 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_ActiveTxDataLanes	R/W	0x1560	0x1	Valid Range: 0x1 - 0x2

PA_ConnectedTxDataLanes	R/W	0x1561	—	Valid Range: 0x0 - 0x2 This value is 0x02 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxTrailingClocks	R/W	0x1564	—	Valid Range: 0x00 - 0xFF (255clocks) This value is 0xFF before Link-up, and is updated during Link-up depending on the capability of host and device
PA_TxPWRStatus	R	0x1567	—	0x0: OFF_STATE 0x1: FAST_STATE 0x2: SLOW_STATE 0x3: HIBERNATE_STATE 0x4: SLEEP_STATE
PA_TxGear	R/W	0x1568	0x1	0x1: PWM_G1, HS_G1 0x2: PWM_G2, HS_G2 0x3: PWM_G3, HS_G3 0x4: PWM_G4, HS_G4
PA_TxTermination	R/W	0x1569	0x00	0: Un-terminated, 1: Terminated
PA_HSSeries ^{Note 3}	R/W	0x156A	0x1	0x1: Rate A Series, 0x2: Rate B Series
PA_PWRMode	R/W	0x1571	0x55	Bit[7:4] RX Power Mode 0x1: Fast_Mode 0x2: Slow_Mode 0x4: FastAuto_Mode 0x5: SlowAuto_Mode 0x7: UNCHANGED
				Bit[3:0] TX Power Mode 0x1: Fast_Mode 0x2: Slow_Mode 0x4: FastAuto_Mode 0x5: SlowAuto_Mode 0x7: UNCHANGED
PA_ActiveRxDataLanes	R/W	0x1580	0x1	Valid Range: 0x1 - 0x2
PA_ConnectedRxDataLanes	R/W	0x1581	—	Valid Range: 0x0 - 0x2 This value is 0x00 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_RxPWRStatus	R	0x1582	—	0x0: OFF_STATE 0x1: FAST_STATE 0x2: SLOW_STATE 0x3: HIBERNATE_STATE 0x4: SLEEP_STATE
PA_RxGear	R/W	0x1583	0x1	0x1: PWM_G1, HS_G1 0x2: PWM_G2, HS_G2 0x3: PWM_G3, HS_G3 0x4: PWM_G4, HS_G4
PA_RxTermination	R/W	0x1584	0x0	0: Un-terminated, 1: Terminated
PA_Scrambling	R/W	0x1585	0x0	
PA_MaxRxPWMGear	R	0x1586	—	This value is 0x01 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_MaxRxHSGear	R	0x1587	—	This value is 0x00 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_PACPRReqTimeout	R/W	0x1590	0x3F	Valid range: 1 - 63 [ms]
PA_PACPRReqEoBTimeout	R/W	0x1591	0x0F	Valid range: 1 - 15 [ms]
PA_RemoteVerInfo	R	0x15A0	—	Undefined before link startup, after link startup depends on peer version information
PA_LogicalLaneMap	R/W	0x15A1	—	Logical lane 0 is mapped to physical lane 0 (TX/RX)
PA_SleepNoConfigTime	R/W	0x15A2	—	Valid range: 0x1 (1SI) - 0xF (15SI) This value is 0x0F before Link-up, and is updated during Link-up depending on the capability of host and device

PA_StallNoConfigTime	R/W	0x15A3	—	Valid range: 0x01 (1SI) - 0xFF (255SI) This value is 0xFF before Link-up, and is updated during Link-up depending on the capability of host and device
PA_SaveConfigTime	R/W	0x15A4	—	Valid range: 0x01 (40ns) - 0xFA (10μs) This value is 0xFA before Link-up, and is updated during Link-up depending on the capability of host and device
PA_RxHSUnterminationCapability	R/W	0x15A5	—	0: FALSE, 1: TRUE This value is 0x0 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_RxLSTerminationCapability	R/W	0x15A6	—	0: FALSE, 1: TRUE This value is 0x0 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_Hibern8Time	R/W	0x15A7	—	Valid range: 0 - 10000 This value is 0x80 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_Tactivate	R/W	0x15A8	—	Valid range: 10 - 10000 This value is 0x10 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_LocalVerInfo	R/W	0x15A9	0x05	Local version info This value is 0x0000 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_Granularity	R/W	0x15AA	—	0x1: 1μs, 0x2: 4μs, 0x3: 8μs, 0x4: 16μs, 0x5: 32μs, 0x6: 100μs This value is 0x06 before Link-up, and is updated during Link-up depending on the capability of host and device
PA_MK2ExtensionGuardBand	R/W	0x15AB	0x0	Valid range: 0 - 10000
PA_PWRModeUserData[0 to11]	R/W	0x15B0 to 0x15BB	All 0x0000	Data to be sent within PACP_PWR_req and delivered to the remote DME Valid range: 0x0000 - 0xFFFF
PA_PACPFrameCount	R/W	0x15C0	—	Valid range: 0x00000000 - 0xFFFFFFFF This counter value is incremented each time valid PACP frame is received
PA_PACPErrorCount	R/W	0x15C1	0x00000000	Valid range: 0x00000000 - 0xFFFFFFFF This counter value is incremented each time erroneous PACP frame is received
PA_PHYTestControl	R/W	0x15C2	0x00	PHY Test Feature control register
				Bit[4] TestPatternSelect: Valid range: 0 - 1
				Bit[3] TestPatternTransmit: Valid range: 0 - 1
				Bit[2] LineReset: Valid range: 0 - 1
				Bit[1] CfgReady: Valid range: 0 - 1
				Bit[0] ContBurst: Valid range: 0 - 1
PA_TxHsG4SyncLength	R/W	0x15D0	—	Start of Burst: High Speed Synchronization pattern Length in HS_G4
PA_TxHsG4PrepareLength	R/W	0x15D1	—	Start of Burst: Minimum Prepare time in HS_G4
PA_PeerRxHsAdaptRefresh	R/W	0x15D2	—	Start of Burst: High Speed adaption pattern Length
PA_PeerRxHsAdaptInitial	R/W	0x15D3	—	Start of Burst: High Speed adaption pattern Length
PA_TxHsAdaptType	R/W	0x15D4	0x03	Adapt Type selector 00 = REFRESH 01 = INITIAL 10 = Reserved 11 = No ADAPT

PA_TxHsAdaptTypeInPA_INIT	R/W	0x15D5	0x02	Adapt Type selector 00 = REFRESH 01 = INITIAL 10 = Previous ADAPT value 11 = No ADAPT
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Note 1: All writable Attributes are reset by power failure or H/W reset. All writable Attributes except for PA_PACPFramCount and PA_PACPErrorCount are reset by EndPointReset or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: PA_HSSeries can be changed only when PA_PWRMode is Slow_Mode or SlowAuto_Mode.

3.5.2. UniPro^R L2 Attributes

Table 17 UniPro^R L2 Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
DL_TxPreemptionCap	R	0x2000	0x01	Supports preemption
DL_TC0TxMaxSDUSize	R	0x2001	0x90	TC0 Tx maximum SDU size is 144 symbols
DL_TC0RxInitCreditVal	R	0x2002	—	
DL_TC1TxMaxSDUSize	R	0x2003	0x90	TC1 Tx maximum SDU size is 144 symbols
DL_TC1RxInitCreditVal	R	0x2004	—	
DL_TC0TxBufferSize	R	0x2005	—	
DL_TC1TxBufferSize	R	0x2006	—	
DL_TC0TXFCThreshold	R/W	0x2040	0x09	Threshold is 9 x 32Bytes Valid range: 0x01 - (DL_PeerTC0RxInitCreditVal - 1)
DL_FC0ProtectionTimeOutVal	R/W	0x2041	0x1FFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_TC0ReplayTimeOutVal	R/W	0x2042	0xFFFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_AFC0ReqTimeOutVal	R/W	0x2043	0x7FFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_AFC0CreditThreshold	R/W	0x2044	0x00	Valid range: 0x00 - 0x1F
DL_TC0OutAckThreshold	R/W	0x2045	0x0	Valid range: 0x0 - 0xF (15Frames)
DL_PeerTC0Present ^{Note 3}	R	0x2046	—	
DL_PeerTC0RxInitCreditVal ^{Note 3}	R	0x2047	—	
DL_TC1TXFCThreshold	R/W	0x2060	0x09	TC1 is not supported
DL_FC1ProtectionTimeOutVal	R/W	0x2061	0x1FFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_TC1ReplayTimeOutVal	R/W	0x2062	0xFFFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_AFC1ReqTimeOutVal	R/W	0x2063	0x7FFF	Valid range: 0x0000 - 0xFFFF (65535μs)
DL_AFC1CreditThreshold	R/W	0x2064	0x00	TC1 is not supported
DL_TC1OutAckThreshold	R/W	0x2065	0x0	Valid range: 0x0 - 0xF (15Frames)
DL_PeerTC1Present ^{Note 3}	R	0x2066	—	
DL_PeerTC1RxInitCreditVal ^{Note 3}	R	0x2067	—	

Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: The value depends on host capability.

3.5.3. UniPro^R L3 Attributes

Table 18 UniPro^R L3 Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
N_DeviceID	R/W	0x3000	0x01	Local Device ID is 0x01
N_DeviceID_valid	R/W	0x3001	0x01	1: TRUE
N_TC0TxMaxSDUSize	R	0x3020	0x0111	Maximum size is 273Bytes
N_TC1TxMaxSDUSize	R	0x3021	0x0111	Maximum size is 273Bytes

Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

3.5.4. UniPro^R L4 Attributes

Table 19 UniPro^R L4 Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset Value ^{Note 2}	Notes
T_NumCPorts	R	0x4000	0x01	1 CPort
T_NumTestFeatures	R	0x4001	0x01	1 Test Feature
T_ConnectionState	R/W	0x4020	0x1	1: Connected
T_PeerDeviceID	R/W	0x4021	0x00	DeviceID of the peer device (host) is 0
T_PeerCPortID	R/W	0x4022	0x00	CPortID of the peer CPort is 0
T_TrafficClass	R/W	0x4023	0x0	Traffic class is 0
T_ProtocolID	R/W	0x4024	0x0000	Reserved for future use
T_CPortFlags	R/W	0x4025	0x6	CPort control register: b0: End-to-End Flow Control (E2EFC) b1: Controlled Segment Dropping (CSD_n) b2: CPort Safety Valve (CSV_n)
T_TxTokenValue ^{Note 3}	R/W	0x4026	0x20	Value of a E2E FC Token transmitted is 32Bytes Valid range: 0x20 - 0x1000000 (powers of 2 only)
T_RxTokenValue ^{Note 3}	R/W	0x4027	0x20	Value of a E2E FC Token received is 32Bytes Valid range: 0x20 - 0x1000000 (powers of 2 only)
T_LocalBufferSpace ^{Note 3}	R/W	0x4028	0x00000000	
T_PeerBufferSpace ^{Note 3}	R/W	0x4029	0x00000000	
T_CreditsToSend ^{Note 3}	R/W	0x402A	0x00000000	
T_CPortMode	R/W	0x402B	0x1	0x1: CPORT_APPLICATION 0x2: CPORT_UNDER_TEST
T_TC0TxMaxSDUSize	R	0x4060	0x110	Maximum transmit payload (SDU) size per Segment for TC0 is 272Bytes
T_TC1TxMaxSDUSize	R	0x4061	0x110	Maximum transmit payload (SDU) size per Segment for TC1 is 272Bytes

Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: These values are invalid because UFS specification doesn't support End-to-End flow control. Don't change these values.

3.5.5. UniPro^R DME DDB L1 Attributes

Table 20 UniPro^R DME DDB L1 Attributes

Attribute Name	Access	Attribute ID	Reset ^{Note 1} Value	Notes
DME_DDBL1_Revision	R	0x5000	0x10	DDB v1.0
DME_DDBL1_Level	R	0x5001	0x01	DDB L2 is not supported
DME_DDBL1_DeviceClass	R	0x5002	0x0002	The Device Class ID of the DME User
DME_DDBL1_ManufacturerID	R	0x5003	0x040E	Manufacturer ID of the DME User
DME_DDBL1_ProductID	R	0x5004	0x0001	The Manufacturer's internal product ID of the DME User is 0
DME_DDBL1_Length	R	0x5005	0x0000	The length of any DDB Level 2 data is 0

Note 1: Reset value means the value after Link-up.

3.5.6. UniPro^R TstSrc / TstDst Attributes

Table 21 UniPro^R TstSrc / TstDst Attributes

Attribute Name	Access ^{Note 1}	Attribute ID	Reset ^{Note 2} Value	Notes
T_TstCPortID	R/W	0x4080	0x00	The ID of the CPort-under-test is 0
T_TstSrcOn	R/W	0x4081	0x00	0: FALSE, 1: TRUE
T_TstSrcPattern	R/W	0x4082	0x00	0: Saw tooth
T_TstSrcIncrement	R/W	0x4083	0x01	Increment value is 1 Valid range: 0x00 - 0xFF
T_TstSrcMessageSize	R/W	0x4084	0x0100	The size of each generated Message Valid range: 0x0001 - 0xFFFF (65535Bytes)
T_TstSrcMessageCount	R/W	0x4085	0x0100	The number of Messages generated Valid range: 0x0000 - 0xFFFF (65535Messages)
T_TstSrcInterMessageGap	R/W	0x4086	0x0000	If non-zero, the gap time between two Messages Valid range: 0x0000 - 0xFFFF (65535μs)
T_TstDstOn	R/W	0x40A1	0x00	0: FALSE, 1: TRUE
T_TstDstErrorDetectionEnable	R/W	0x40A2	0x00	0: FALSE, 1: TRUE
T_TstDstPattern	R/W	0x40A3	0x00	0: Saw tooth
T_TstDstIncrement	R/W	0x40A4	0x01	Increment value is 1 Valid range: 0x00 - 0xFF
T_TstDstMessageCount	R/W	0x40A5	0x0000	Number of received Messages Valid range: 0x0000 - 0xFFFF (65535Messages)
T_TstDstMessageOffset	R/W	0x40A6	0x0000	Offset of error from start of Message Valid range: 0x0000 - 0xFFFF (65535Bytes)
T_TstDstMessageSize	R/W	0x40A7	0x0100	The expected size of incoming Message Valid range: 0x0000 - 0xFFFF (65535Bytes)
T_TstDstFCCredits ^{Note 3}	R/W	0x40A8	0x0000	Valid range: 0x1 - 0xFFFFFFFF (4294967295Bytes)
T_TstDstInterFCTokenGap ^{Note 3}	R/W	0x40A9	0x0000	Valid range: 0x0000 - 0xFFFF (65535μs)
T_TstDstInitialFCCredits ^{Note 3}	R/W	0x40AA	0x0000	Valid range: 0x1 - 0xFFFFFFFF (4294967295Bytes)

T_TstDstErrorCode	R/W	0x40AB	0x00	0x0: NO_ERROR 0x1: FRAGMENT_CORRUPT 0x2: INVALID_MSG_SIZE 0x3: UNEXPECTED_BYTE_VALUE
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Note 1: All writable Attributes are reset by power failure, H/W reset, EndPointReset, or Link lost.

Note 2: Reset value means the value after Link-up.

Note 3: These values are invalid because UFS specification doesn't support End-to-End flow control. Don't change these values.

3.6. UPIU Transaction codes

THGJFGT1E45BAIP supports following UPIU transaction codes. Each command is used with corresponding UPIU. It is not allowed to use undefined codes. Refer to UFS specification for more detailed information.

Table 22 UPIU Transaction Codes

Initiator To Target	Transaction Code	Target to Initiator	Transaction Code
NOP Out	0x00	NOP In	0x20
Command	0x01	Response	0x21
Data Out	0x02	Data In	0x22
Task Management Request	0x04	Task Management Response	0x24
Reserved	0x11	Ready To Transfer	0x31
Query Request	0x16	Query Response	0x36
Reserved	0x1F	Reject UPIU	0x3F

3.7. UFS Descriptors

The UFS modules support following UFS descriptors defined in UFS specification. Please refer to UFS specification for more detailed information.

Table 23 UFS Descriptor TYPE and IDN

DESCRIPTOR TYPE	DESCRIPTOR IDN	DESCRIPTOR TYPE	DESCRIPTOR IDN
DEVICE	0x00	STRING	0x05
CONFIGURATION	0x01	RFU	0x06
UNIT	0x02	GEOMETRY	0x07
RFU	0x03	POWER	0x08
INTERCONNECT	0x04	DEVICE HEALTH	0x09

3.7.1. Device Descriptor

Table 24 Device Descriptor

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLength	0x59	—	Size of this descriptor
0x01	1	bDescriptorIDN	0x00	—	Device Descriptor Type Identifier
0x02	1	bDevice	0x00	—	0x00: Device
0x03	1	bDeviceClass	0x00	—	0x00: Mass Storage
0x04	1	bDeviceSubClass	0x00	—	0x00: Embedded Bootable
0x05	1	bProtocol	0x00	—	0x00: SCSI
0x06	1	bNumberLU	0x00	YES	Valid range: 0x00 - 0x1F (Default value: 0x00)
0x07	1	bNumberWLU	0x04	—	0x04: 4 Well known logical Units exist
0x08	1	bBootEnable	0x00	YES	0x00: Boot feature disabled (Default) 0x01: Bootable feature enabled
0x09	1	bDescrAccessEn	0x00	YES	0x00: Device Descriptor access disabled (Default) 0x01: Device Descriptor access enabled
0x0A	1	bInitPowerMode	0x01	YES	0x00: UFS-Sleep Mode 0x01: Active Mode (Default)
0x0B	1	bHighPriorityLUN	0x7F	YES	0x00 - 0x1F: Configured LUN has high priority 0x7F: All logical units have the same priority (Default)
0x0C	1	bSecureRemovalType	0x00	YES	Secure Removal Type Valid range: 0x00 (Default), 0x03
0x0D	1	bSecurityLU	0x01	—	0x01: RPMB
0x0E	1	bBackgroundOpsTermLat	0x0F	—	0x0F: 150ms
0x0F	1	bInitActiveICCLLevel	0x00	YES	bInitActiveICCLLevel defines the bActiveICCLLevel value after power on or reset Valid range: 0x00 - 0x0F (Default value: 0x00)
0x10	2	wSpecVersion	0x0310	—	Specification version
0x12	2	wManufactureDate	—	—	Manufacturing Date
0x14	1	iManufacturerName	0x01	—	Manufacturer Name
0x15	1	iProductName	0x02	—	Product Name
0x16	1	iSerialNumber	0x03	—	Serial Number
0x17	1	iOemID	0x04	—	OEM ID
0x18	2	wManufacturerID	0x0198	—	Manufacturer ID
0x1A	1	bUD0BaseOffset	0x16	—	Unit Descriptor 0 Base Offset
0x1B	1	bUDConfigPLength	0x1A	—	Unit Descr. Config. Param. Length Total size of the configurable Unit Descriptor parameters
0x1C	1	bDeviceRTTCap	0x02	—	RTT Capability of device
0x1D	2	wPeriodicRTCUpdate	0x0000	YES	Frequency and method of Real-Time Clock update (Default value: 0x0000)
0x1F	1	bUFSFeaturesSupport	0xB3	—	
0x20	1	bFFUTimeout	0x01	—	
0x21	1	bQueueDepth	0x40	—	Shared Queue Depth = 64
0x22	2	wDeviceVersion	0x0000	—	
0x24	1	bNumSecureWPArea	0x20	—	
0x25	4	dPSAMaxDataSize ^{Note 1}	0x00000000	—	max. 0x00100000
0x29	1	bPSAStateTimeout	0x0D	—	819.2ms
0x2A	1	iProductRevisionLevel	0x05	—	
0x2B	21	Reserved	—	—	
0x40	2	wHPBVersion	0x0200	—	HPB v2.0

0x42	1	bHPBControl	0x01	YES	0x00: Host control mode 0x01: Device control mode
0x43	12	Reserved	—	—	
0x4F	4	dExtendedUFSFeaturesSupport	0x000001B3	—	Extended UFS features support Bit[0]: Field Firmware Update (FFU) Bit[1]: Production State Awareness (PSA) Bit[2]: Device Life Span Bit[3]: Refresh Operation (only UFS3.0) Bit[4]: TOO_HIGH_TEMPERATURE Bit[5]: TOO_LOW_TEMPERATURE Bit[6]: Extended Temperature Bit[7]: Host Performance Booster (HPB) Bit[8]: WriteBooster Bit[9]: Performance Throttling Bit[31:10]: Reserved
0x53	1	bWriteBoosterBufferPreserveUserSpaceEn	0x00	YES	Default value: 0x00 Only User space is not reduced is supported (0x01) 0x00: User space is reduced if WriteBooster Buffer is configured 0x01: User space is not reduced if WriteBooster Buffer is configured
0x54	1	bWriteBoosterBufferType	0x00	YES	Default value: 0x00 Only Single Shared WriteBuffer type (0x01) is supported 0x00: LU based WriteBooster Buffer configuring 0x01: Single shared WriteBooster Buffer configuring
0x55	4	dNumSharedWriteBoosterBufferAllocUnits	0x00000000	YES	The WriteBooster Buffer size for a shared WriteBooster Buffer configuration Value expressed in unit of Allocation Units If this value is '0', then the shared WriteBooster is not configured for the device

Note 1: bPSAMaxDataSize value is updated automatically by the device after device configuration.

3.7.2. Configuration Descriptor [Index0]

Table 25 Configuration Descriptor [Index0]

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLength	0xE6	—	Size of this descriptor
0x01	1	bDescriptorIDN	0x01	—	Configuration Descriptor Type Identifier
0x02	1	bConfDescContinue	0x00	YES	
0x03	1	bBootEnable	0x00	YES	Boot Enable
0x04	1	bDescrAccessEn	0x00	YES	Descriptor Access Enable
0x05	1	bInitPowerMode	0x01	YES	Initial Power Mode
0x06	1	bHighPriorityLUN	0x7F	YES	High Priority LUN
0x07	1	bSecureRemovalType	0x00	YES	Secure Removal Type
0x08	1	bInitActiveICCLLevel	0x00	YES	Initial Active ICC Level
0x09	2	wPeriodicRTCUpdate	0x0000	YES	Frequency and method of Real-Time Clock update
0x0B	1	bHPBControl	0x01	YES	0x00: Host control mode 0x01: Device control mode
0x0C	1	bRPMBRegionEnable	0x00	YES	RPMB Region Enable
0x0D	1	bRPMBRegion1Size	0x00	YES	RPMB Region 1 Size
0x0E	1	bRPMBRegion2Size	0x00	YES	RPMB Region 2 Size
0x0F	1	bRPMBRegion3Size	0x00	YES	RPMB Region 3 Size
0x10	1	bWriteBoosterBufferPreserveUserSpaceEn	0x00	YES	0x00: User Space reduction mode (not supported) 0x01: No User Space reduction mode

0x11	1	bWriteBoosterBufferType	0x00	YES	WriteBooster Buffer Type 0x00: LU dedicated buffer type (not supported) 0x01: Single shared buffer type
0x12	4	dNumSharedWriteBoosterBufferAllocUnits	0x00000000	YES	The WriteBooster Buffer size for the Logical Unit

3.7.3. Configuration Descriptor [Index1-3]

Table 26 Configuration Descriptor [Index1-3]

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLength	0xE6	—	Size of this descriptor
0x01	1	bDescriptorIDN	0x01	—	Configuration Descriptor Type Identifier
0x02	1	bConfDescContinue	0x00	—	
0x03	19	Reserved	—	—	

3.7.4. Unit Descriptor configurable parameters

Table 27 Unit Descriptor configurable parameters

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLUEnable	0x00	YES	Logical Unit Enable
0x01	1	bBootLunID	0x00	YES	Boot LUN ID
0x02	1	bLUWriteProtect	0x00	YES	Logical Unit Write Protect
0x03	1	bMemoryType	0x00	YES	Memory Type
0x04	4	dNumAllocUnits	0x00000000	YES	Number of Allocation Units
0x08	1	bDataReliability	0x00	YES	Data Reliability
0x09	1	bLogicalBlockSize	0x0C	YES	Logical Block Size
0x0A	1	bProvisioningType	0x00	YES	Provisioning Type
0x0B	2	wContextCapabilities	0x0000	YES	Context Capabilities
0x0D	3	Reserved	—	—	
0x10	2	wLUMaxActiveHPBRegions	0x0000	YES	max. number of HPB Regions assigned to the HPB logical unit
0x12	2	wHPBPinnedRegionStartIdx	0x0000	YES	HPB Pinned Region Start Offset
0x14	2	wNumHPBPinnedRegions	0x0000	YES	Number of HPB pinned Regions assigned to the HPB logical unit
0x16	4	dLUNumWriteBoosterBufferAllocUnits	0x00000000	YES	Logical Unit Maximum WriteBoosterSize

3.7.5. Geometry Descriptor

Table 28 Geometry Descriptor

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x57	Byte of device geometry information
0x01	1	bDescriptorIDN	0x07	Geometry Descriptor Type Identifier
0x02	1	bMediaTechnology	0x00	Reserved
0x03	1	Reserved	—	

0x04	8	qTotalRawDeviceCapacity	0x00000000_1DCBC000	Total Raw Device Capacity
0x0C	1	bMaxNumberLU	0x01	32 Logical Units
0x0D	4	dSegmentSize	0x00002000	4MBytes
0x11	1	bAllocationUnitSize	0x01	Value expressed in number of Segments
0x12	1	bMinAddrBlockSize	0x08	4KBytes
0x13	1	bOptimalReadBlockSize	0x80	64KBytes
0x14	1	bOptimalWriteBlockSize	0x08	4KBytes
0x15	1	bMaxInBufferSize	0x08	4KBytes
0x16	1	bMaxOutBufferSize	0x40	32KBytes
0x17	1	bRPMB_ReadWriteSize	0xF8	Maximum 62K of RPMB frames allowed in Security Protocol In/Out
0x18	1	bDynamicCapacityResourcePolicy	0x01	
0x19	1	bDataOrdering	0x00	0x00: out-of-order data transfer is not supported by the device
0x1A	1	bMaxContextIDNumber	0x05	Maximum 5 contexts are available
0x1B	1	bSysDataTagUnitSize	0x00	Tag Unit Size is 4KBytes
0x1C	1	bSysDataTagResSize	0x00	System Data Tag Resource Size is 0.1%
0x1D	1	bSupportedSecRTypes	0x09	Supported Secure Removal Types
0x1E	2	wSupportedMemoryTypes	0x800F	Supported Memory Types: Normal memory type System code memory type Non-Persistent memory type Enhanced memory type 1 RPMB memory type
0x20	4	dSystemCodeMaxNAllocU ^{Note 1}	0x0000EE5E	max. Number of Allocation Units for the System Code memory type
0x24	2	wSystemCodeCapAdjFac	0x0300	Capacity Adjustment Factor for the System Code memory type
0x26	4	dNonPersistMaxNAllocU ^{Note 1}	0x0000EE5E	max. Number of Allocation Units for the Non-Persistent memory type
0x2A	2	wNonPersistCapAdjFac	0x0300	Capacity Adjustment Factor for the Non-Persistent memory type
0x2C	4	dEnhanced1MaxNAllocU ^{Note 1}	0x0000EE5E	max. Number of Allocation Units for the Enhanced memory type 1
0x30	2	wEnhanced1CapAdjFac	0x0300	Capacity Adjustment Factor for the Enhanced memory type 1
0x32	4	dEnhanced2MaxNAllocU	0x00000000	Enhanced memory type 2 is not supported
0x36	2	wEnhanced2CapAdjFac	0x0000	
0x38	4	dEnhanced3MaxNAllocU	0x00000000	Enhanced memory type 3 is not supported
0x3C	2	wEnhanced3CapAdjFac	0x0000	
0x3E	4	dEnhanced4MaxNAllocU	0x00000000	Enhanced memory type 4 is not supported
0x42	2	wEnhanced4CapAdjFac	0x0000	
0x44	4	dOptimalLogicalBlockSize	0x00000000	4KBytes
0x48	1	bHPBRegionSize	0x0F	HPB Region Size: 16MB
0x49	1	bHPBNumberLU	0x02	max. number of HPB LU: 2
0x4A	1	bHPBSubRegionSize	0x0F	HPB Sub-Region Size: 16MB
0x4B	2	wDeviceMaxActiveHPBRegions	0x2000	max. number of Active HPB Regions: 8192
0x4D	2	Reserved	—	
0x4F	4	dWriteBoosterBufferMaxNAllocUnits	0x00002000	Maximum total WriteBooster Buffer size
0x53	1	bDeviceMaxWriteBoosterLUs	0x01	Number of maximum WriteBooster Buffer

0x54	1	bWriteBoosterBufferCapAdjFac	0x03	Capacity Adjustment Factor for the WriteBooster Buffer memory type 0x02: for MLC NAND 0x03: for TLC NAND
0x55	1	bSupportedWriteBoosterBuffer UserSpaceReductionTypes	0x01	0x00: WriteBooster buffer can be configured only in user space reduction type 0x01: WriteBooster buffer can be configured only in user space is not reduction type 0x02: Device can be configured in either user space reduction type or no user space reduction type
0x56	1	bSupportedWriteBoosterBuffer Types	0x01	0x01: Single shared WriteBooster Buffer configuring (Default value: 0x01)

Note 1: If WriteBooster Buffer is configured, regardless of max. number of Allocation unit, the total size of dEnhanced1MaxNAllocU, dSystemCodeMaxNAllocU and dNonPersistMaxNAllocU which can be used is limited to 1GB.

3.7.6. Unit Descriptor

THGJFGT1E45BAIP has 32 Unit Descriptors. All default values except for bUnitIndex are the same.

Table 29 Unit Descriptor

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLength	0x2D	—	Size of this descriptor
0x01	1	bDescriptorIDN	0x02	—	Unit Descriptor Type Identifier
0x02	1	bUnitIndex	0x00 to 0x1F	—	Unit Index
0x03	1	bLUEnable	0x00	YES	0x00: Logical Unit disabled (Default) 0x01: Logical Unit enabled 0x02: Host Performance Booster (HPB)
0x04	1	bBootLunID	0x00	YES	0x00: Not bootable (Default) 0x01: Boot LU A 0x02: Boot LU B
0x05	1	bLUWriteProtect	0x00	YES	0x00: LU not write protected (Default) 0x01: LU write protected when fPowerOnWPEn = 1 0x02: LU permanently write protected when fPermanentWPEn = 1
0x06	1	bLUQueueDepth	0x00	—	
0x07	1	bPSASensitive ^{Note 1}	0x00	—	0x00: Enhanced memory type1 (Default) 0x01: Normal Memory Type
0x08	1	bMemoryType	0x00	YES	0x00: Normal Memory (Default) 0x01: System code memory type 0x02: Non-Persistent memory type 0x03: Enhanced memory type 1
0x09	1	bDataReliability	0x00	YES	0x00: The logical unit is not protected (Default) 0x01: The logical unit is protected
0x0A	1	bLogicalBlockSize	0x0C	YES	Value range: 0x0C (Default) - 0x0Fh
0x0B	8	qLogicalBlockCount	0x00000000_00000000	YES	This value can be configured by setting the dNumAllocUnits parameter of the Configuration Descriptor Default value: 0x00000000_00000000
0x13	4	dEraseBlockSize	0x00000000	—	Erase Block Size
0x17	1	bProvisioningType	0x00	YES	0x00: Thin Provisioning is disabled (Default) 0x02: Thin Provisioning is enabled and TPRZ = 0 0x03: Thin Provisioning is enabled and TPRZ = 1
0x18	8	qPhyMemResourceCount	0x00000000_00000000	—	qPhyMemResourceCount shall be equal to qLogicalBlockCount initially

0x20	2	wContextCapabilities	0x0000	YES	Bit[6:4]: LARGE_UNIT_MAX_MULTIPLIER_M1 Default value: b000
					Bit[3:0]: MaxContextID Default value: b0000
0x22	1	bLargeUnitGranularity_M1	0x00	—	Large Unit Granularity = 1MB x (bLargeUnitGranularity_M1 + 1)
0x23	2	wLUMaxActiveHPBRegions	0x0000	YES	Maximum Number of Active HPB Regions
0x25	2	wHPBPinnedRegionStartIdx	0x0000	YES	HPB Pinned Region Start Offset
0x27	2	wNumHPBPinnedRegions	0x0000	YES	Number of HPB Pinned Regions
0x29	4	dLUNumWriteBoosterBuffer AllocUnits	0x00000000	YES	Not support LU-based WriteBooster Configuration

Note 1: bPSASensitive value is updated automatically by the device after device configuration.

3.7.7. RPMB Unit Descriptor

Table 30 RPMB Unit Descriptor

OFFSET	SIZE	NAME	MDV	User Conf.	DESCRIPTION
0x00	1	bLength	0x23	—	Size of this descriptor
0x01	1	bDescriptorIDN	0x02	—	Unit Descriptor Type Identifier
0x02	1	bUnitIndex	0xC4	—	Unit Index
0x03	1	bLUEnable	0x01	—	0x01: Logical Unit enabled
0x04	1	bBootLunID	0x00	—	0x00: Not bootable
0x05	1	bLUWriteProtect	0x00	—	0x00: LU not write protected
0x06	1	bLUQueueDepth	0x01	—	Logical Unit Queue Depth
0x07	1	bPSASensitive	0x00	—	
0x08	1	bMemoryType	0x0F	—	0x0F: RPMB Memory Type
0x09	1	bRPMBRegionEnable	0x00	YES	RPMB Region
0x0A	1	bLogicalBlockSize	0x08	—	0x08: 256Bytes Logical Block Size
0x0B	8	qLogicalBlockCount	0x00000000 _00010000	—	16MB
0x13	1	bRPMBRegion0Size	0x80	YES	RPMB Region 0 Size 16MB
0x14	1	bRPMBRegion1Size	0x00	YES	RPMB Region 1 Size
0x15	1	bRPMBRegion2Size	0x00	YES	RPMB Region 2 Size
0x16	1	bRPMBRegion3Size	0x00	YES	RPMB Region 3 Size
0x17	1	bProvisioningType	0x00	—	0x00: Thin Provisioning is disabled
0x18	8	qPhyMemResourceCount	0x00000000 _00010000	—	16MB
0x20	3	Reserved	—	—	

3.7.8. Power Parameters Descriptor

Table 31 Power Parameters Descriptor

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x62	Size of this descriptor
0x01	1	bDescriptorIDN	0x08	Power Parameters Descriptor Type Identifier

0x02	2	wActiveICCLevelsVCC (0)	0x8190	2Bytes maximum V _{CC} current value for each of the 16 active current consumption levels starting with level 0
0x04	2	wActiveICCLevelsVCC (1)	0x8190	
0x06	2	wActiveICCLevelsVCC (2)	0x8190	
0x08	2	wActiveICCLevelsVCC (3)	0x8190	
0x0A	2	wActiveICCLevelsVCC (4)	0x8190	
0x0C	2	wActiveICCLevelsVCC (5)	0x8190	
0x0E	2	wActiveICCLevelsVCC (6)	0x8190	
0x10	2	wActiveICCLevelsVCC (7)	0x8190	
0x12	2	wActiveICCLevelsVCC (8)	0x8190	
0x14	2	wActiveICCLevelsVCC (9)	0x8190	
0x16	2	wActiveICCLevelsVCC (10)	0x8190	
0x18	2	wActiveICCLevelsVCC (11)	0x8190	
0x1A	2	wActiveICCLevelsVCC (12)	0x8190	
0x1C	2	wActiveICCLevelsVCC (13)	0x8190	
0x1E	2	wActiveICCLevelsVCC (14)	0x8190	
0x20	2	wActiveICCLevelsVCC (15)	0x8190	
0x22	2	wActiveICCLevelsVCCQ (0)	0x81F4	2Bytes maximum V _{CCQ} current value for each of the 16 active current consumption levels starting with level 0
0x24	2	wActiveICCLevelsVCCQ (1)	0x81F4	
0x26	2	wActiveICCLevelsVCCQ (2)	0x81F4	
0x28	2	wActiveICCLevelsVCCQ (3)	0x81F4	
0x2A	2	wActiveICCLevelsVCCQ (4)	0x81F4	
0x2C	2	wActiveICCLevelsVCCQ (5)	0x81F4	
0x2E	2	wActiveICCLevelsVCCQ (6)	0x81F4	
0x30	2	wActiveICCLevelsVCCQ (7)	0x81F4	
0x32	2	wActiveICCLevelsVCCQ (8)	0x81F4	
0x34	2	wActiveICCLevelsVCCQ (9)	0x81F4	
0x36	2	wActiveICCLevelsVCCQ (10)	0x81F4	
0x38	2	wActiveICCLevelsVCCQ (11)	0x81F4	
0x3A	2	wActiveICCLevelsVCCQ (12)	0x81F4	
0x3C	2	wActiveICCLevelsVCCQ (13)	0x81F4	
0x3E	2	wActiveICCLevelsVCCQ (14)	0x81F4	
0x40	2	wActiveICCLevelsVCCQ (15)	0x81F4	
0x42	2	wActiveICCLevelsVCCQ2 (0)	0x0000	2Bytes maximum V _{CCQ2} current value for each of the 16 active current consumption levels starting with level 0
0x44	2	wActiveICCLevelsVCCQ2 (1)	0x0000	
0x46	2	wActiveICCLevelsVCCQ2 (2)	0x0000	
0x48	2	wActiveICCLevelsVCCQ2 (3)	0x0000	
0x4A	2	wActiveICCLevelsVCCQ2 (4)	0x0000	
0x4C	2	wActiveICCLevelsVCCQ2 (5)	0x0000	
0x4E	2	wActiveICCLevelsVCCQ2 (6)	0x0000	
0x50	2	wActiveICCLevelsVCCQ2 (7)	0x0000	
0x52	2	wActiveICCLevelsVCCQ2 (8)	0x0000	
0x54	2	wActiveICCLevelsVCCQ2 (9)	0x0000	
0x56	2	wActiveICCLevelsVCCQ2 (10)	0x0000	
0x58	2	wActiveICCLevelsVCCQ2 (11)	0x0000	
0x5A	2	wActiveICCLevelsVCCQ2 (12)	0x0000	
0x5C	2	wActiveICCLevelsVCCQ2 (13)	0x0000	
0x5E	2	wActiveICCLevelsVCCQ2 (14)	0x0000	
0x60	2	wActiveICCLevelsVCCQ2 (15)	0x0000	

3.7.9. Device Health Descriptor

Table 32 Device Health Descriptor

OFFSET	SIZE	NAME	Value	DESCRIPTION
0x00	1	bLength	0x2D	Size of this descriptor
0x01	1	bDescriptorIDN	0x09	Device Health Descriptor Type Identifier
0x02	1	bPreEOLInfo	0x01	Pre End of Life Information This field provides indication about device life time reflected by average reserved blocks 0x00: Not defined 0x01: Normal (Default) 0x02: Warning 0x03: Critical Others: Reserved
0x03	1	bDeviceLifeTimeEstA	0x01	This field provides an indication of the device life time based on the amount of performed program / erase cycles. The calculation method is vendor specific and referred as method A. 0x00: Not defined 0x01: 0% - 10% device life time used (Default) 0x02: 10% - 20% device life time used 0x03: 20% - 30% device life time used 0x04: 30% - 40% device life time used 0x05: 40% - 50% device life time used 0x06: 50% - 60% device life time used 0x07: 60% - 70% device life time used 0x08: 70% - 80% device life time used 0x09: 80% - 90% device life time used 0x0A: 90% - 100% device life time used 0x0B: Exceeded its maximum estimated device life time Others: Reserved
0x04	1	bDeviceLifeTimeEstB	0x01	This field provides an indication of the device life time based on the amount of performed program / erase cycles. The calculation method is vendor specific and referred as method B. 0x00: Not defined 0x01: 0% - 10% device life time used (Default) 0x02: 10% - 20% device life time used 0x03: 20% - 30% device life time used 0x04: 30% - 40% device life time used 0x05: 40% - 50% device life time used 0x06: 50% - 60% device life time used 0x07: 60% - 70% device life time used 0x08: 70% - 80% device life time used 0x09: 80% - 90% device life time used 0x0A: 90% - 100% device life time used 0x0B: Exceeded its maximum estimated device life time Others: Reserved
0x05	31	Reserved	—	
0x24	1	bDeviceLifetimeEstC	0x00	This field provides an indication of the device life time based on the amount of performed program / erase cycles. It represents the majority of NAND dies such as TLC with 1% granularity. 00h: 0% 01h: 1% device life time used 02h: 2% device life time used : 63h: 99% device life time used 64h: 100% device life time used
0x25	4	dRefreshTotalCount	—	Refresh specified by JEDEC is not supported
0x29	4	dRefreshProgress	—	

3.7.10. Interconnect Descriptor**Table 33 Interconnect Descriptor**

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x06	Size of this descriptor
0x01	1	bDescriptorIDN	0x04	Interconnect Descriptor Type Identifier
0x02	2	bcdUniproVersion	0x0180	MIPI Unipro ^R Release Version Number
0x04	2	bcdMphyVersion	0x0410	MIPI M-PHY ^R Release Version Number

3.7.11. Manufacturer Name String Descriptor**Table 34 Manufacture Name String Descriptor**

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x12	Size of this descriptor
0x01	1	bDescriptorIDN	0x05	String Descriptor Type Identifier
0x02	2	UC	0x004B	Unicode string character ("K")
0x04	2	UC	0x0049	Unicode string character ("I")
0x06	2	UC	0x004F	Unicode string character ("O")
0x08	2	UC	0x0058	Unicode string character ("X")
0x0A	2	UC	0x0049	Unicode string character ("I")
0x0C	2	UC	0x0041	Unicode string character ("A")
0x0E	2	UC	0x0020	Unicode string character (" ")
0x10	2	UC	0x0020	Unicode string character (" ")

3.7.12. Product Name String Descriptor**Table 35 Product Name String Descriptor**

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x22	Size of this descriptor
0x01	1	bDescriptorIDN	0x05	String Descriptor Type Identifier
0x02	2	UC	0x0054	Unicode string character ("T")
0x04	2	UC	0x0048	Unicode string character ("H")
0x06	2	UC	0x0047	Unicode string character ("G")
0x08	2	UC	0x004A	Unicode string character ("J")
0x0A	2	UC	0x0046	Unicode string character ("F")
0x0C	2	UC	0x0047	Unicode string character ("G")
0x0E	2	UC	0x0054	Unicode string character ("T")
0x10	2	UC	0x0031	Unicode string character ("1")
0x12	2	UC	0x0045	Unicode string character ("E")
0x14	2	UC	0x0034	Unicode string character ("4")
0x16	2	UC	0x0035	Unicode string character ("5")
0x18	2	UC	0x0042	Unicode string character ("B")
0x1A	2	UC	0x0041	Unicode string character ("A")
0x1C	2	UC	0x0049	Unicode string character ("I")
0x1E	2	UC	0x0050	Unicode string character ("P")
0x20	2	UC	0x0042	Unicode string character ("B")

3.7.13. OEM ID String Descriptor

Table 36 OEM ID String Descriptor

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0xFE	Size of this descriptor
0x01	1	bDescriptorIDN	0x05	String Descriptor Type Identifier
0x02	252	UC	0x00	Unicode string character Data size is (bLength-2)

3.7.14. Serial Number String Descriptor

Table 37 Serial Number String Descriptor

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x16	Size of this descriptor
0x01	1	bDescriptorIDN	0x05	String Descriptor Type Identifier
0x02	20	UC[0-9]	— Note 1	Unicode string character Valid range: 0x0030 - 0x0039, 0x0041 - 0x005A

Note 1: This Memory supports 10 UNICODE characters unique serial number of each device.

3.7.15. Product Revision Level String Descriptor

Table 38 Product Revision Level String Descriptor

OFFSET	SIZE	NAME	VALUE	DESCRIPTION
0x00	1	bLength	0x0A	Size of this descriptor
0x01	1	bDescriptorIDN	0x05	String Descriptor Type Identifier
0x02	2	UC	0x0030	Unicode string character ("0100")
0x04	2	UC	0x0031	
0x06	2	UC	0x0030	
0x08	2	UC	0x0030	

3.8. UFS Flags

Table 39 UFS Flags

IDN	NAME	TYPE	DEFAULT VALUE	DESCRIPTION
0x00	Reserved	—	—	
0x01	fDeviceInit	Read / Set Only	0	0: Device initialization completed or not started yet 1: Device initialization in progress
0x02	fPermanentWPEn	Read / Write Once	0	0: Permanent write protection disabled 1: Permanent write protection enabled
0x03	fPowerOnWPEn	Read / Power on reset	0	0: Power on write protection disabled 1: Power on write protection enabled
0x04	fBackgroundOpsEn	Read / Volatile	1	0: Device initiated background ops. is disabled 1: Device initiated background ops. is enabled
0x05	fDeviceLifeSpanModeEn	Read / Volatile	0	0: Device Life Span Mode is disabled
0x06	fPurgeEnable	Write Only / Volatile	—	0: Purge operation is disabled 1: Purge operation is enabled
0x07	fRefreshEnable	Write Only / Volatile	0	0: Forced Refresh operation is disabled 1: Forced Refresh operation is enabled
0x08	fPhyResourceRemoval	Read / Persistent	0	Physical Resource Removal
0x09	fBusyRTC	Read Only	0	0: Device is not executing internal operation related to RTC
0x0A	Reserved	—	—	
0x0B	fPermanentlyDisableFwUpdate	Read / Write Once	0	0: The UFS device firmware may be modified 1: The UFS device shall permanently disallow future firmware updates to the UFS device
0x0C	Reserved	—	—	
0x0D	Reserved	—	—	
0x0E	fWriteBoosterEn	Read / Volatile	0	WriteBooster Enable 0: WriteBooster is not enabled 1: WriteBooster is enabled
0x0F	fWriteBoosterBufferFlushEn	Read / Volatile	0	Flush the data in WriteBooster buffer to the user area of storage 0: Flush operation is not performed 1: Flush operation is performed
0x10	fWriteBoosterBufferFlushDuringHibernateEnter	Read / Volatile	0	Flush WriteBooster buffer during Hibernation state 0: Device is not allowed to flush the WriteBooster buffer during Hibernation state of the device 1: Device is allowed to flush the WriteBooster buffer during Hibernation state of the device
0x11	fHPBReset	Read / Set Only	0	HPB Reset 0: HPB reset completed or not started yet 1: HPB reset in progress
0x12	fHPBEn	Read / Persistent	0	HPB Enable 0: HPB is not enabled 1: HPB is enabled

3.9. UFS Attributes

Table 40 UFS Attributes

IDN	NAME	TYPE	SIZE [Byte]	MDV	DESCRIPTION
0x00	bBootLunEn	Read / Persistent	1	0x00	0x00: Boot disabled 0x01: Enabled boot from Boot LU A 0x02: Enabled boot from Boot LU B
0x01	bMAX_DATA_SIZE_FOR_HPB_SINGLE_CMD	Read / Read Only	1	0x07	The size is calculated as (bMAX_DATA_SIZE_FOR_HPB_SINGLE_CMD + 1) x 4KB
0x02	bCurrentPowerMode ^{Note 1}	Read Only	1	Note 1	Current Power Mode
0x03	bActiveICCLLevel	Read / Volatile	1	0x00	Active ICC Level
0x04	bOutOfOrderDataEn	Read / Write Once	1	0x00	0x00: Out-of-order data transfer is disabled
0x05	bBackgroundOpStatus	Read Only	1	0x00	Background Operations Status
0x06	bPurgeStatus	Read Only	1	0x00	Purge Operation Status
0x07	bMaxDataInSize	Read / Persistent	1	0x08	512Bytes x 8 = 4KBytes
0x08	bMaxDataOutSize	Read / Persistent	1	0x40	512Bytes x 40 = 32KBytes
0x09	dDynCapNeeded	Read Only	4	0x00000000	Dynamic Capacity Needed
0x0A	bRefClkFreq	Read / Persistent	1	0x01	0x00: 19.2MHz 0x01: 26MHz 0x02: 38.4MHz
0x0B	bConfigDescrLock	Read / Write Once	1	0x00	0x00: Configuration Descriptor and OEM ID String Descriptor not locked 0x01: Configuration Descriptor and OEM ID String Descriptor locked
0x0C	bMaxNumOfRTT	Read / Persistent	1	0x02	0x02: Maximum 2 RTTs
0x0D	wExceptionEventControl	Read / Volatile	2	0x0000	Bit[6]: PERFORMANCE_THROTTLING_EN
					Bit[5]: WRITEBOOSTER_EVENT_EN
					Bit[4]: TOO_LOW_TEMP_EN
					Bit[3]: TOO_HIGH_TEMP_EN
					Bit[2]: URGENT_BKOPS_EN
					Bit[1]: SYSPPOOL_EVENT_EN
					Bit[0]: DYNCAP_EVENT_EN
0x0E	wExceptionEventStatus	Read Only	2	0x0000	Bit[6]: PERFORMANCE_THROTTLING
					Bit[5]: WRITEBOOSTER_FLUSH_NEEDED
					Bit[4]: TOO_LOW_TEMP
					Bit[3]: TOO_HIGH_TEMP
					Bit[2]: URGENT_BKOPS
					Bit[1]: SYSPPOOL_EXHAUSTED
					Bit[0]: DYNCAP_NEEDED
0x0F	dSecondsPassed	Write Only	4	0x00000000	Bit[31:0]: Seconds passed from TIME BASELINE
0x10	wContextConf	Read / Volatile	2	0x0000	INDEX specifies the LU number. SELECTOR specifies the Context ID within the LU
0x11	Obsolete	—	—	—	
0x12	Reserved	—	—	—	
0x13	Reserved	—	—	—	

0x14	bDeviceFFUStatus	Read Only	1	0x00	Device FFU Status 0x00: No information 0x01: Successful microcode update 0x02: Microcode corruption error 0x03: Internal error 0x04: Microcode version mismatch 0x05 - 0xFE: Reserved 0xFF: General Error
0x15	bPSAState ^{Note 2}	Read / Persistent	1	0x00	Product State Awareness State 0x00: Off 0x01: Pre-soldering 0x02: Loading Complete 0x03: Soldered
0x16	dPSADDataSize	Read / Persistent	4	0x00000000	The amount of data that the host plans to load to all logical units with bPSASensitive set to 1
0x17	bRefClkGatingWaitTime	Read Only	1	0x0F	Minimum time for which the reference clock is required by device during transition to LS-MODE or HIBERN8 state
0x18	bDeviceCaseRoughTemperaure	Read Only	1	0x00	Device's rough package case surface temperature
0x19	bDeviceTooHighTempBoundary	Read Only	1	0xBE	High temperature boundary
0x1A	bDeviceTooLowTempBoundary	Read Only	1	0x32	Low temperature boundary
0x1B	bThrottlingStatus	Read Only	1	0x00	Result in performance throttling Bit[0]: Temperature
0x1C	bWriteBoosterBufferFlushStatus	Read Only	1	0x00	Flush operation status of WriteBooster buffer 0x00: idle (no flush operation) 0x01: Flush operation in progress 0x02: Flush operation stopped prematurely by host 0x03: Flush operation completed successfully 0x04: Flush operation general failure
0x1D	bAvailableWriteBoosterBufferSize	Read Only	1	0x00	Available WriteBooster Buffer Size 0x00: 0% buffer remains 0x01 - 0x09: 10% - 90% buffer remain 0x0A: 100% buffer remains
0x1E	bWriteBoosterBufferLifeTimeEst	Read Only	1	0x00	This field provides an indication of the WriteBooster Buffer life time based on the amount of performed program / erase cycles 0x00: Information not available 0x01: 0% - 10% device life time used 0x02: 10% - 20% device life time used ⋮ 0x08: 70% - 80% device life time used 0x09: 80% - 90% device life time used 0x0A: 90% - 100% device life time used
0x1F	dCurrentWriteBoosterBufferSize	Read Only	4	0x00000000	Current WriteBooster Buffer Size
0x20-0x2B	Reserved	—	—	—	
0x2C	bRefreshStatus	Read Only	1	0x00	Refresh specified by JEDEC is not supported
0x2D	bRefreshFreq	Read / Persistent	1	0x00	
0x2E	bRefreshUnit	Read / Persistent	1	0x00	
0x2F	bRefreshMethod	Read / Write Once	1	0x00	

Note 1: bCurrentPowerMode value after initialization changes depending on the value of bInitPowerMode.

Note 2: It is prohibited to use 4KB size read request to verify preloaded data during Pre-soldering state or Loading Complete state. If 4KB read request is sent by accident, host should ignore HPB Active region information from device.

3.10. UFS SCSI Commands

Table 41 UFS SCSI Commands

Command name	Opcode	Command name	Opcode
FORMAT UNIT	0x04	SECURITY PROTOCOL IN	0xA2
INQUIRY	0x12	SECURITY PROTOCOL OUT	0xB5
MODE SELECT (10)	0x55	SEND DIAGNOSTIC	0x1D
MODE SENSE (10)	0x5A	START STOP UNIT	0x1B
PRE-FETCH (10)	0x34	SYNCHRONIZE CACHE (10)	0x35
PRE-FETCH (16)	0x90	SYNCHRONIZE CACHE (16)	0x91
READ (6)	0x08	TEST UNIT READY	0x00
READ (10)	0x28	UNMAP	0x42
READ (16)	0x88	VERIFY (10)	0x2F
READ BUFFER	0x3C	WRITE (6)	0x0A
READ CAPACITY (10)	0x25	WRITE (10)	0x2A
READ CAPACITY (16)	0x9E	WRITE (16)	0x8A
REPORT LUNS	0xA0	WRITE BUFFER	0x3B
REQUEST SENSE	0x03	HPB READ	0xF8, 0x88 ^{Note 1}
HPB READ BUFFER	0xF9	HPB WRITE BUFFER	0xFA

Note 1: Bit5 to 7 of Byte1 in CDB format of HPB Read command changes from Reserved to 111b to distinguish existing command and HPB read command when OPERATION CODE (0x88) is used.

3.11. Mode Pages

3.11.1. Control Mode Page

Table 42 Control Mode Page

Bit	7	6	5	4	3	2	1	0
Byte								
0	PS = 1b	SPF = 0b	PAGE CODE = 001010b					
1	PAGE LENGTH = 0x0A							
2	TST = 000b			TMF_ONLY = 0b	DPICZ = 0b	D_SENSE = 0b	GLTSD = 0b	RLEC = 0b
3	QUEUE ALGORITHM MODIFIER = 0001b				NUAR = 0b	QERR = 00b		Obsolete = 0b
4	VS = 0b	RAC = 0b	UA_INTLCK_CTRL = 00b		SWP = 0b	Obsolete = 000b		
5	ATO = 0b	TAS = 0b	ATMPE = 0b	RWWP = 0b	Reserved = 0b	AUTOLOAD MODE = 000b		
6	Obsolete = 0x0000							
7								
8	BUSY TIMEOUT PERIOD = 0xFFFF							
9								
10	EXTENDED SELF-TEST COMPLETION TIME = 0x0000							
11								

3.11.2. Read-Write Error Recovery Mode Page

Table 43 Read-Write Error Recovery Mode Page

Bit Byte	7	6	5	4	3	2	1	0
0	PS = 0b	SPF = 0b	PAGE CODE = 000001b					
1	PAGE LENGTH = 0x0A							
2	AWRE = 1b	ARRE = 0b	TB = 0b	RC = 0b	EER = 0b	PER = 0b	DTE = 0b	DCR = 0b
3	READ RETRY COUNT = 0x00							
4	Obsolete = 0x00							
5	Obsolete = 0x00							
6	Obsolete = 0x00							
7	TPERE = 0b	Reserved = 00000b					Restricted for MMC-6 = 00b	
8	WRITE RETRY COUNT = 0x00							
9	Reserved = 0x00							
10	(MSB)							
11	RECOVERY TIME LIMIT = 0xFFFF (LSB)							

3.11.3. Caching Mode Page

Table 44 Caching Mode Page

Bit Byte	7	6	5	4	3	2	1	0	
0	PS = 1b	SPF = 0b	PAGE CODE = 01000b						
1	PAGE LENGTH = 0x12								
2	IC = 0b	ABPF = 0b	CAP = 0b	DISC = 0b	SIZE = 0b	WCE = 1b	MF = 0b	RCD = 0b	
3	DEMAND READ RETENTION PRIORITY = 0000b				WRITE RETENTION PRIORITY = 0000b				
4	(MSB)	DISABLE PRE-FETCH TRANSFER LENGTH = 0x0000							(LSB)
5									
6	(MSB)	MINIMUM PRE-FETCH = 0x0000							(LSB)
7									
8	(MSB)	MAXIMUM PRE-FETCH = 0x0000							(LSB)
9									
10	(MSB)	MAXIMUM PRE-FETCH CEILING = 0x0000							(LSB)
11									
12	FSW = 0b	LBCSS = 0b	DRA = 0b	Vendor Specific = 00b		Reserved = 00b		NV_DIS = 0b	
13	NUMBER OF CACHE SEGMENTS = 0x00								
14	(MSB)	CACHE SEGMENT SIZE = 0x0000							(LSB)
15									

16	Reserved = 0x00
17	
18	Obsolete = 0x00 0000
19	

3.12. Standard INQUIRY data

Table 45 Standard INQUIRY data format

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE 00000b ^{Note 1} 11110b ^{Note 2}				
1	RMB = 0b	Reserved = 000000b						
2	VERSION = 0x06							
3	Obsolete = 0b	Obsolete = 0b	NORMACA = 0b	HISUP = 0b	RESPONSE DATA FORMAT = 0010b			
4	ADDITIONAL LENGTH (n-4) = 0x1F							
5	SCCS = 0b	ACC = 0b	TPGS = 00b		3PC = 0b	Reserved = 00b		PROTECT = 0b
6	Obsolete = 0b	ENC SERV = 0b	VS = 0b	MULTIP = 0b	Obsolete = 0b	Obsolete = 0b	Obsolete = 0b	ADDR16 = 0b
7	Obsolete = 0b	Obsolete = 0b	WBUS16 = 0b	SYNC = 0b	Obsolete = 0b	Obsolete = 0b	CMDQUE = 1b	VS = 0b
8	(MSB) _____							
15	VENDOR IDENTIFICATION = 0x4B494F58 49412020 _____ (LSB)							
16	(MSB) _____							
31	PRODUCT IDENTIFICATION (Refer to Table 46) _____ (LSB)							
32	(MSB) _____							
35	PRODUCT REVISION LEVEL = 0x30313030 (0100) _____ (LSB)							

Note 1: Direct Access Device for logical unit (non well known)

Note 2: Well known logical unit

Table 46 PRODUCT IDENTIFICATION

Part Number	Density	PRODUCT IDENTIFICATION
THGJFGT1E45BAIP	256GB	0x5448474A 46475431 45343542 41495042 (THGJFGT1E45BAIPB)

3.13. Vital Product Data

Table 47 Supported Vital Product Data

Vital Product Data Name	Page Code	Well Known LU	Except Well Known LU
Supported VPD Pages	0x00	Supported	Supported
Device Identification	0x83	Supported	Supported
Mode Page Policy	0x87	Not supported	Supported
Unit Serial Number	0x80	Supported	Supported
Block Limits	0xB0	Not supported	Supported
Block Device Characteristics	0xB1	Supported	Supported
Logical Block Provisioning	0xB2	Not supported	Supported

3.13.1. Supported VPD Pages

Table 48 Supported VPD Pages (Well Known LU)

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE = 11110b				
1	PAGE CODE = 0x00							
2	PAGE LENGTH = 0x0004							
3								
4	Supported VPD Pages = 0x00							
5	Unit Serial Number = 0x80							
6	Device Identification = 0x83							
7	Block Device Characteristics VPD page = 0xB1							

Table 49 Supported VPD Pages (Except Well Known LU)

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE = 00000b				
1	PAGE CODE = 0x00							
2	PAGE LENGTH = 0x0007							
3								
4	Supported VPD Pages = 0x00							
5	Unit Serial Number = 0x80							
6	Device Identification = 0x83							
7	Mode Page Policy = 0x87							
8	Block Limits VPD page = 0xB0							
9	Block Device Characteristics VPD page = 0xB1							
10	Logical Block Provisioning VPD page = 0xB2							

3.13.2. Device Identification

Table 50 Device Identification

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE 00000b Note 1 11110b Note 2				
1	PAGE CODE = 0x83							
2	PAGE LENGTH = 0x000C							
3								
4	PROTOCOL IDENTIFIER = 0x0				CODE SET = 0x1			
5	PIV = 0b	Reserved = 0b	ASSOCIATION 00b Note 1 10b Note 2		DESIGNATOR TYPE = 0010b			
6	Reserved = 0x00							
7	DESIGNATOR LENGTH = 0x08							
8	IEEE COMPANY_ID = 0x8CE38E							
9								
10								
11	VENDOR SPECIFIC EXTENSION IDENTIFIER = 0x0000000000							
12								
13								
14								
15								

Note 1: Direct Access Device for logical unit (non well known)

Note 2: Well known logical unit

3.13.3. Mode Page Policy

Table 51 Mode Page Policy

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE = 00000b				
1	PAGE CODE = 0x87							
2	PAGE LENGTH = 0x000C							
3								
4	Reserved = 00b		POLICY PAGE CODE = 001010b					
5	POLICY SUB PAGE CODE = 0x00							
6	MLUS = 0b	Reserved = 00000b					MODE PAGE POLICY = 00b	

7	Reserved = 0x00		
8	Reserved = 00b	POLICY PAGE CODE = 000001b	
9	POLICY SUB PAGE CODE = 0x00		
10	MLUS = 1b	Reserved = 00000b	MODE PAGE POLICY = 00b
11	Reserved = 0x00		
12	Reserved = 00b	POLICY PAGE CODE = 001000b	
13	POLICY SUB PAGE CODE = 0x00		
14	MLUS = 1b	Reserved = 00000b	MODE PAGE POLICY = 00b
15	Reserved = 0x00		

3.13.4. Unit Serial Number

Table 52 Unit Serial Number

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE 00000b Note 1 11110b Note 2				
1	PAGE CODE = 0x80							
2	PAGE LENGTH = 0x000A							
3								
4	PRODUCT SERIAL NUMBER[0]							
5	PRODUCT SERIAL NUMBER[1]							
6	PRODUCT SERIAL NUMBER[2]							
7	PRODUCT SERIAL NUMBER[3]							
8	PRODUCT SERIAL NUMBER[4]							
9	PRODUCT SERIAL NUMBER[5]							
10	PRODUCT SERIAL NUMBER[6]							
11	PRODUCT SERIAL NUMBER[7]							
12	PRODUCT SERIAL NUMBER[8]							
13	PRODUCT SERIAL NUMBER[9]							

Note 1: Direct Access Device for logical unit (non well known)

Note 2: Well known logical unit

3.13.5. Block Limits

Table 53 Block Limits

Bit	7	6	5	4	3	2	1	0
Byte								
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE = 00000b				
1	PAGE CODE = 0xB0							
2	PAGE LENGTH = 0x003C							
3								
4	Reserved = 0x00							
5	MAXIMUM COMPARE AND WRITE LENGTH = 0x00							
6	OPTIMAL TRANSFER LENGTH GRANULARITY = 0x0001 *In case of Logical Block Size is 4KB							
7								
8	MAXIMUM TRANSFER LENGTH = 0x0000 0000							
:								
11								
12								
:	OPTIMAL TRANSFER LENGTH = 0x0000 0080 *In case of Logical Block Size is 512KB							
15								
16								
:	MAXIMUM PREFETCH XDREAD XDWRITE TRANSFER LENGTH = 0x0000 0000							
19								
20								
:	MAXIMUM UNMAP LBA COUNT = 0xFFFF FFFF							
23								
24								
:	MAXIMUM UNMAP BLOCK DESCRIPTOR COUNT = 0x0000 0001							
27								
28								
:	OPTIMAL UNMAP GRANULARITY = 0x0000 0001							
31								
32	UGAVALID = 0b	UNMAP GRANULARITY ALIGNMENT = 0000000b						
33	UNMAP GRANULARITY ALIGNMENT = 0x00 0000							
:								
35								
36	Reserved = all 0x00							
:								
63								

3.13.6. Block Device Characteristics

Table 54 Block Device Characteristics

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE 00000b Note 1 11110b Note 2				
1	PAGE CODE = 0xB1							
2	PAGE LENGTH = 0x003C							
3								
4	MEDIUM ROTATION RATE = 0x0001							
5								
6	Reserved = 0x00							
7	Reserved = 0000b				NOMINAL FORM FACTOR = 0000b			
8	Reserved = all 0x00							
⋮								
63								

Note 1: Direct Access Device for logical unit (non well known)

Note 2: Well known logical unit

3.13.7. Logical Block Provisioning

Table 55 Logical Block Provisioning VPD

Bit Byte	7	6	5	4	3	2	1	0
0	PERIPHERAL QUALIFIER = 000b			PERIPHERAL DEVICE TYPE = 00000b				
1	PAGE CODE = 0xB2							
2	PAGE LENGTH = 0x0004							
3								
4	THRESHOLD EXPONENT ^{Note 2}							
5	TPU ^{Note 1}	TPWS = 0b	Reserved = 00b		ANC_SUP = 000b			DP = 0b
6	Reserved = 0x0000							
7								

Note 1: The value of TPU is dependent on the configuration of logical units. If the logical unit is configured as thin provisioning, the value is 1. If the logical unit is configured as full provisioning, the value is 0.

Note 2: The value of THRESHOLD EXPONENT is dependent on the configuration of logical units.

4. ELECTRICAL CHARACTERISTICS

4.1. Absolute Maximum Ratings

The absolute maximum ratings of a semiconductor device are a set of specified parameter values, which must not be exceeded during operation, even for an instant. If any of these rating would be exceeded during operation, the electrical characteristics of the device may be irreparably altered and the reliability and lifetime of the device can no longer be guaranteed. Moreover, these operations with exceeded ratings may cause break down, damage, and/or degradation to any other equipment. Applications using the device should be designed such that each maximum rating will never be exceeded in any operating conditions. Before using, creating, and/or producing designs, refer to and comply with the precautions and conditions set forth in this document.

Table 56 Absolute Maximum Ratings

Parameter	Symbol	min.	max.	Unit
Supply voltage 1	V_{CC}	-0.3	3.9	V
Supply voltage 2	V_{CCQ}	-0.2	1.6	V
Input Voltage (M-PHY ^R Signals, REF_CLK, RST_n)	V_{IO}	-0.2	1.6	V

4.2. H/W Reset Operation

Table 57 RST_n Characteristics

Parameter	Symbol	min.	max.	Unit
Input High voltage ^{Note 1}	V_{IH}	$0.65 \times V_{CCQ}$	$V_{CCQ} + 0.3$	V
Input Low voltage ^{Note 1}	V_{IL}	$V_{SS} - 0.3$	$0.35 \times V_{CCQ}$	V
Input Capacitance	C_{in}	—	10	pF
Input leakage Current	I_{lk}	—	10	μA
Pulse Width	t_{RSTW}	1	—	μs
High Period (Interval)	t_{RSTH}	1	—	μs
Filtering Period	t_{RSTF}	100	—	ns

Note 1: Users need to input V_{CCQ} (1.14 to 1.26V) based signals to RST_n to meet the electrical characteristics defined by UFS specification.

4.3. Reference Clock

Table 58 REF_CLK Characteristics

Parameter	Symbol	min.	max.	Unit
Frequency	f_{ref}	19.2 26 38.4		MHz
Frequency Error	f_{ERROR}	-150	+150	ppm
Input High Voltage ^{Note 1}	V_{IH}	$0.65 \times V_{\text{CCQ}}$	—	V
Input Low Voltage ^{Note 1}	V_{IL}	—	$0.35 \times V_{\text{CCQ}}$	V
Clock Rise Time ^{Note 2}	t_{RISE}	—	2	ns
Clock Fall Time ^{Note 2}	t_{FALL}	—	2	ns
Duty Cycle	t_{DC}	45	55	%
Phase Noise	N	—	-66	dBc
Noise Floor Density	N_{density}	—	-140	dBc/Hz
Input impedance	RL_{RX}	100	—	k Ω
	CL_{RX}	—	5	pF

Note 1: Users need to input V_{CCQ} (1.14 to 1.26V) based signals to REF_CLK to meet the electrical characteristics defined by UFS specification.

Note 2: Clock rise time and clock fall time shall be measured from 20% to 80% of the window defined by $V_{\text{IL}}(\text{max.})$ to $V_{\text{IH}}(\text{min.})$.

4.4. Power Supply

4.4.1. Power Supply Voltage

Table 59 Supply Voltage

Parameter	Symbol	min.	max.	Unit
Supply voltage 1	V_{CC} ^{Note 1}	2.7	3.6	V
		2.4	2.7	V
Supply voltage 2	V_{CCQ} ^{Note 1}	1.14	1.26	V
Supply voltage power-up timing for V_{CC}	t_{PRUH}	0.01	35	ms
Supply voltage power-up timing for V_{CCQ}	t_{PRUV}	0.01	20	ms

Note 1: It is required to design layouts that supply above V_{CC} and V_{CCQ} voltage at the solder ball input terminals.

4.4.2. Power up/down Sequence

Users should follow power up/down sequence defined in JEDEC UFS specification Ver.3.1 and this document. UFS device requires the following sequence during power-up in addition to JEDEC UFS specification.

4.4.2.1. Power Up Sequence

- Timing definition of t_{PRUH} and t_{PRUV} .

t_{PRUH} and t_{PRUV} are defined as the time from when V_{CC} and V_{CCQ} exceed 0.3V till when V_{CC} and V_{CCQ} reach minimum operation voltage, $V_{CC}(\text{min.})$ and $V_{CCQ}(\text{min.})$. t_{PRUH} and t_{PRUV} need to satisfy the values shown at Table 59.

- Timing definition after supply voltages exceed a specified values

V_{CC} and V_{CCQ} should reach valid voltage range within 0.5ms after exceeding 75% of $V_{CC}(\text{min.})$ and $V_{CCQ}(\text{min.})$.

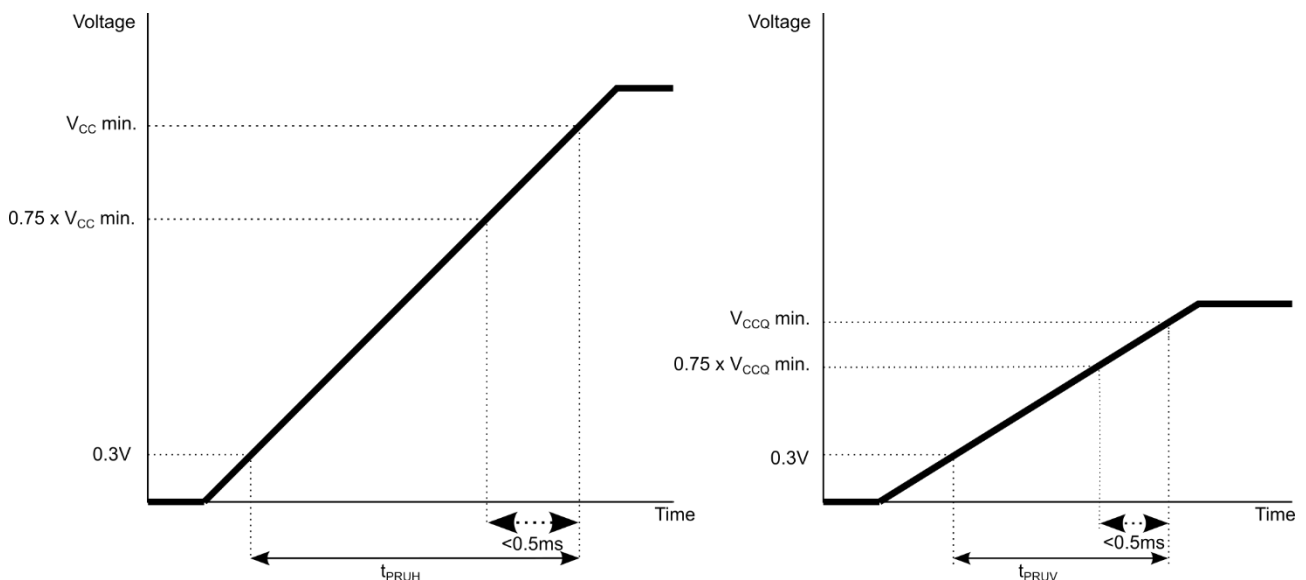


Figure 7 Power up sequence

4.4.2.2. Power Down Sequence

- Once the power supply V_{CC} or V_{CCQ} falls below the minimum guaranteed voltage (for example, upon sudden power fail), the voltage level of V_{CC} or V_{CCQ} shall be kept less than 0.3V for at least 1ms before it goes beyond 0.3V again.
- Supply voltage shall be designed to avoid faster power down curve than $-100\text{mV}/\mu\text{s}$ in all cases.

4.4.3. Operating Current (RMS)

Table 60 Maximum Supply Current

Parameter	HS-Gear	No. of Lane	Symbol	Current	Unit
Maximum Operating Current	HS-G4	2	I_{CC}	405	mA
			I_{CCQ}	580	mA

Note 1: The measurement for max. RMS current is done as average RMS current consumption over a period of 100ms.

Note 2: The conditions of maximum values are 85°C , $V_{CC} = 3.6\text{V}$, $V_{CCQ} = 1.26\text{V}$.

4.4.4. Sleep Mode Current and Timing Definition

The device can transit to UFS Sleep / M-PHY^R Hibern8 by issuing Start Stop Unit command and Hibern8 Enter command. Table 61 shows the current consumption during UFS Sleep / M-PHY^R Hibern8 and Table 62 shows the timing definition that the device transit to the status after Hibern8 Enter command is accepted in the device.

Table 61 Sleep Mode Current

Parameter	Symbol	typ. ^{Note 1}	max. ^{Note 2}	Unit
Sleep Mode Current	I_{CCQS}	470	9700	μA
	$I_{CCS} + I_{CCQS}$	590	10100	μA

Note 1: The conditions of typical values are 25°C , $V_{CC} = 2.5\text{V}$, $V_{CCQ} = 1.2\text{V}$, UFS Sleep / M-PHY^R Hibern8.

Note 2: The conditions of maximum values are 85°C , $V_{CC} = 3.6\text{V}$, $V_{CCQ} = 1.26\text{V}$, UFS Sleep / M-PHY^R Hibern8.

Table 62 Sleep Mode Transition Timing

Parameter	Symbol	min.	max.	Unit
Transition timing to UFS Sleep / M-PHY ^R Hibern8	t_{SLEEP}	—	1000	μs

Note: The starting point of transition timing is when a host recognizes HIBERNATE_ENTER.ind. Thus, a host needs to wait for more than t_{SLEEP} time after receiving HIBERNATE_ENTER.ind.

4.5. Power mode control

4.5.1. Unipro^R power mode control

4.5.1.1. Restriction of reference clock

- Stopping reference clock in entering Hibern8
A host shall drive reference clock for 15 μ s (bRefClkGatingWaitTime) after detecting HibernEnter.ind.
- Re-starting reference clock in exiting Hibern8
A host shall drive reference clock before requesting HibernExit.req.
- Wait for 1ms after receiving the response for DME_HIBERNATE_ENTER
A host shall wait for 1ms after receiving the HibernEnter.ind. then turn V_{CC} off or power mode of V_{CCQ} change.

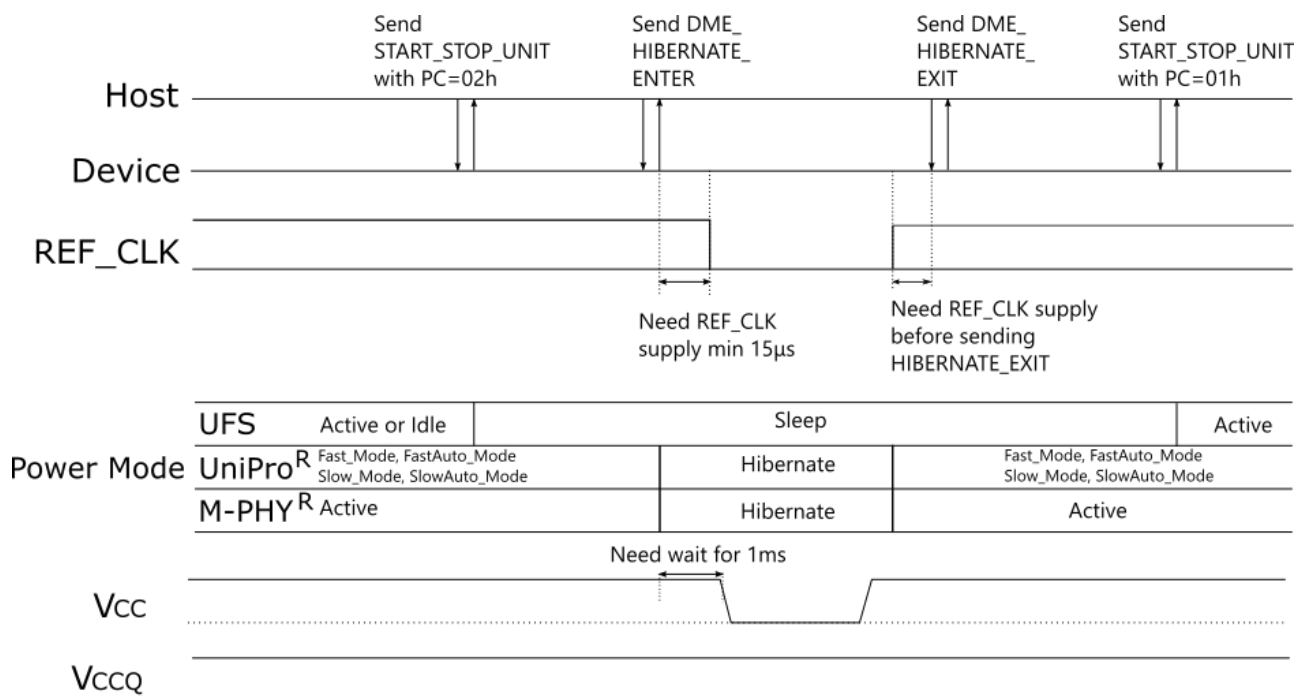


Figure 8 Restriction on reference clock during Power mode change to/from Sleep

5. Reliability Guidance

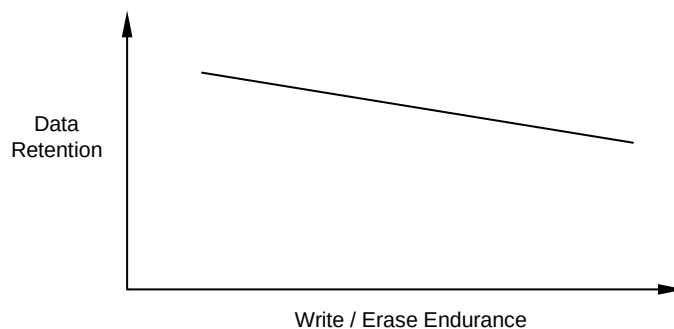
This reliability guidance is intended to notify some guidance related to using raw NAND flash. Although random bit errors may occur during use, it does not necessarily mean that a block is bad. Generally, a block should be marked as bad when a program status failure or erase status failure is detected. The other failure modes may be recovered by a block erase. ECC treatment for read data is mandatory due to the following Data Retention and Read Disturb failures.

- Write / Erase Endurance

Write / Erase endurance failures may occur in a cell, page, or block, and are detected by doing a status read after either an auto program or auto block erase operation. The cumulative bad block count will increase along with the number of write / erase cycles.

- Data Retention

The data in memory may change after a certain amount of storage time. This is due to charge loss or charge gain. After block erasure and reprogramming, the block may become usable again. Also write / erase endurance deteriorates data retention capability. The figure below shows a generic trend of relationship between write / erase endurance and data retention.



- Read Disturb

A read operation may disturb the data in memory. The data may change due to charge gain. Usually, bit errors occur on other pages in the block, not the page being read. After a large number of read cycles (between block erases), a tiny charge may build up and can cause a cell to be soft programmed to another state. After block erasure and reprogramming, the block may become usable again.

Considering the above failure modes, following usages are recommended:

- Please avoid any excessive iteration of resets and initialization sequences (Device identification mode) as far as possible after power-on, which may result in read disturb failure. The resets include hardware resets and software resets including EndPointReset and Link lost.
- Please avoid any excessive iteration of power mode change from Active or Idle power mode to UFS-Sleep or UFS-PowerDown power mode by START_STOP_UNIT. This condition could be preventable by inserting a write operation.

6. Reference

- JEDEC UFS specification Ver. 3.1
- MIPI UniPro^R specification Ver. 1.8
- MIPI M-PHY^R specification Ver. 4.1

7. Document Revision History

Rev.0.10	Aug. 3rd, 2021	Released as preliminary version
Rev.1.00	Sep. 1st, 2021	Released as first revision. Fixed some typos

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